

Quantifying Heterogeneity in Bovine Tuberculosis Breakdown Dynamics Using Clustering and Finite Mixture Modelling

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Abstract

This study applies unsupervised learning techniques to identify distinct epidemiological profiles of bovine tuberculosis (bTB) outbreaks using herd-level data. A set of seven variables capturing breakdown duration, test results, recurrence, and herd characteristics were analysed following standardisation.

Clustering was performed using K-means, Partitioning Around Medoids (PAM), and Gaussian Mixture Models (GMM) to assess robustness across methods. The optimal solution identified three clusters.

Cluster characterisation revealed clear differences in breakdown characteristics. Cluster 3 breakdowns were associated with longer breakdown durations, increased transmission indicators (ie, additional cases in subsequent herd tests), and a greater history of prior breakdowns. Cluster 1 breakdowns exhibited fewer reactors and shorter durations, while Cluster 2 displayed intermediate characteristics. GMM analysis supported the presence of underlying structure but favoured a two-cluster solution, highlighting some uncertainty in boundary cases.

Survival analysis further demonstrated significant differences in time to subsequent breakdown across clusters (log-rank test, $p < 0.0001$). Herds in Cluster 1 remained breakdown-free for longer periods, indicating lower recurrence risk, whereas Cluster 2 showed the fastest decline in survival probability, reflecting a higher likelihood of rapid recurrence. Cluster 3 displayed an intermediate pattern, with a gradual decline over time.

Our findings highlight the heterogeneous nature of bTB breakdowns, which might reflect different infection transmission dynamics. The differences observed in survival to subsequent breakdowns might support more tailored approaches to herd breakdowns.

These findings reinforce the epidemiological relevance of the identified clusters and suggest that they capture meaningful differences in disease persistence and reinfection dynamics.

Keywords: Bovine tuberculosis, Clustering, PAM, Epidemiology, Breakdown Heterogeneity

Contents

1	Introduction	3
1.1	Historical context and current situation regarding bovine tuberculosis (bTB) in Ireland	3
1.2	Motivation for examining heterogeneity in breakdowns	4
1.3	Role of clustering in investigating heterogeneity	5
1.4	Study objectives	5
2	Materials and Methods	5
2.1	Datasets maintained and curated as part of the national programme	5
2.2	Principal Component Analysis	7
2.3	Clustering method	8
2.3.1	Partitioning Around Medoids (PAM)	8
3	Multinomial Regression	8
3.1	Odds Ratios and Confidence Intervals	10
3.1.1	Survival Analysis of Time to Next Breakdown by Cluster	13
3.1.2	Software and reproducibility	13
4	Results	13
4.1	Exploratory Data Analysis	13
4.2	PCA Results	14
4.2.1	Structure of the PCA space	15
4.2.2	Variable contributions to principal components	15
4.2.3	Description of Clusters	17
4.2.4	Mean values of variables by cluster (K-means, $K = 3$)	17
4.2.5	Silhouette comparison of K-means and PAM	18
4.2.6	Description of Cluster Results	18
4.3	Temporal distribution of clusters	19
4.4	Summary of results	21
4.4.1	PAM cluster profiles	21
4.5	Comparison with other clustering methods	21
5	Discussion	23
5.0.1	Kaplan–Meier survival curves	49

1. Introduction

1.1. Historical context and current situation regarding bovine tuberculosis (bTB) in Ireland

Bovine tuberculosis (bTB) caused by **Mycobacterium bovis** represents a persistent challenge for Ireland’s agricultural sector. Eradication matters greatly for Ireland’s standing in international trade, as countries with official tuberculosis-free status gain competitive advantages in selling livestock and dairy products globally [1]. In addition, the disease carries zoonotic risks, as humans can contract tuberculosis from unpasteurized dairy products and direct contact with infected cattle, making TB a genuine occupational hazard for farmers and dairy workers [2].

Ireland operates within a complex European regulatory framework that shapes its bovine tuberculosis control strategy. Cattle vaccination for bTB control remains illegal across the EU, restricting Ireland’s policy options and preventing the adoption of vaccination as a supplementary control measure despite experimental evidence of potential benefits [3]. The European Union’s broader framework for TB in wildlife recognises that bTB persistence in countries like Ireland reflects the presence of wildlife reservoirs and the need for integrated surveillance across domestic and wild animal populations [4]. As a non-TB-free country within the EU, Ireland must maintain rigorous testing and surveillance protocols mandated by EU regulations, creating ongoing financial and administrative burdens that persist until eradication is achieved and official TB-free status is granted [5].

The financial damage caused by bovine tuberculosis extends beyond individual farm losses, affecting entire agricultural systems and national economies. For farmers, the primary economic impact comes from testing costs and the restrictions placed on animal movement when herds are suspected or confirmed as infected. Control of bTB in the Republic of Ireland costs approximately €84 million each year [6], though this official figure likely underestimates the true economic burden. As bTB prevalence decreases, the testing burden becomes increasingly costly because more negative tests are needed to confirm herd freedom, and false positive results lead to unnecessary culling of valuable animals. A double financial blow to farmers already struggling with restrictions. The ripple effects touch multiple sectors: trade partners impose import bans on cattle and dairy products from affected regions, markets shrink, farm productivity drops, and families face mental and financial stress from years of repeated breakdowns and restrictions. Ireland bears these mounting costs year after year without achieving eradication, making it clear that sustained, inadequate control programmes drain resources indefinitely when what the country needs is decisive action to end the disease entirely [7].

Ireland’s persistent bovine tuberculosis challenge reflects three interconnected epidemiological barriers that confound eradication efforts. Wildlife reservoirs, particularly European

badgers, maintain infection within the landscape and serve as a continuous source of transmission to cattle populations, undermining progress from cattle-focused control measures alone [8]. Residual infection where infected cattle remain undetected due to imperfect diagnostic sensitivity creates silent reservoirs within herds that perpetuate transmission cycles and seed recurrent breakdowns [9]. Cattle trade movements introduce infected animals, often undetected carriers, across regions, with genomic evidence showing that inward cattle movements are a primary driver of infection in high-risk facilities [10]. Addressing bTB eradication therefore demands simultaneous intervention across all three routes: enhanced detection of infected cattle, management of badger populations as wildlife reservoirs, and risk-based restrictions on cattle trade to prevent geographic spread [7]. Without such a comprehensive, multi-faceted strategy targeting wildlife, residual infection, and movement simultaneously, Ireland’s national eradication programme will remain constrained [10].

1.2. Motivation for examining heterogeneity in breakdowns

Bovine tuberculosis breakdowns in Irish herds do not follow a single pattern but instead vary considerably in their characteristics [11]. Surveillance records and field observations show that breakdowns have changed over time, with differences in how often they occur, how severe they are, how long they last, and whether they happen again [12]. Some herds experience small outbreaks with only a few infected animals that clear quickly, while others face serious within-herd spread, long restriction periods, or multiple breakdowns in succession [13]. These differences suggest that various underlying causes drive different breakdown types, rather than a single disease process affecting all herds.

Multiple factors working together at different levels explain why breakdowns might differ. Variation in transmission within herds, depending on infection levels and the reliability of tests, affects outbreak size and how long infection persists [11]. Between herds, cattle movements carry disease to new farms, Wildlife interactions and environmental conditions also shape whether infection arrives and how long it stays [14]. Different combinations of these factors create distinct breakdown profiles with different risk characteristics.

From a practical disease control standpoint, understanding different breakdown types might help target resources better. Small, contained outbreaks need different responses than large, persistent ones or herds that break down repeatedly [15]. Better understanding of breakdown patterns would allow the programme to target control measures more precisely and use resources more efficiently [15]. Recognising these differences is essential for improving Ireland’s overall bTB control strategy.

Classifying breakdowns into groups with similar patterns helps move beyond looking only at averages and reveals the real structure in surveillance data. Data-driven methods can sort

breakdowns into meaningful categories based on their actual characteristics without forcing assumptions about what patterns exist [11]. This opens the door to examining trends over time, differences between regions, and specific risk factors driving each breakdown type. Understanding this heterogeneity is a critical foundation for grasping how bTB actually spreads and persists in Irish cattle herds.

1.3. Role of clustering in investigating heterogeneity

Clustering groups bTB breakdowns by their shared characteristics without forcing predetermined categories onto the data [11]. This approach works well when breakdown types have not been clearly defined, allowing researchers to discover patterns naturally present in surveillance records [16]. This sorted classification then supports systematic study of how these breakdown types change over time and vary across different regions [15]. Clustering therefore gives researchers a clear, data-driven way to understand and explain the different forms that bTB takes across Irish herds [17].

1.4. Study objectives

Based on the need to better understand heterogeneity in bovine tuberculosis (bTB) breakdowns, the objectives of this study were:

1. To quantify and characterise heterogeneity in bovine tuberculosis breakdowns using clustering and finite mixture modelling applied to breakdown-level surveillance data.
2. To describe trends in these clusters overtime.
3. To identify risk factors for each cluster type.
4. To quantify survival to a subsequent breakdown depending on cluster type.

2. Materials and Methods

2.1. Datasets maintained and curated as part of the national programme

Data used in this study came from Ireland’s national bovine tuberculosis surveillance systems, which track bTB cases as part of the country’s ongoing eradication programme [1]. These systems collect detailed information on herd breakdowns, including test results, how long each outbreak lasted, and whether herds became infected again [15]. The country also maintains records of cattle movements between herds over time, which helps researchers trace how the disease spreads from one farm to another [2]. Information on which herds are located near each other allows researchers to measure the risk of disease transmission between neighbouring farms [18]. All the data goes through checks to make sure it is complete and accurate before being combined for analysis [19]. Together, these different sources of data give researchers a comprehensive picture of how bTB moves through Irish cattle populations and help guide control decisions [19].

Data Preparation and Variable Selection

Breakdown-Level Filtering

To investigate heterogeneity in bovine tuberculosis (bTB) breakdowns, breakdown-level surveillance records were processed to retain epidemiologically meaningful events suitable for unsupervised clustering analysis.

Breakdowns were excluded if they ÷
had zero detected cases;

had a duration shorter than 60 days, or occurred within a herd designated as a controlled finishing unit (CFU) herds. CFU herds are specialised production systems where cattle may be introduced from restricted herds under intensive finishing conditions. To ensure comparability across epidemiological events, only breakdowns initiated by routine slaughter surveillance or standard single intradermal comparative tuberculin testing (SICTT) were retained.

Construction of Clustering Variables

A set of seven variables was defined to capture key epidemiological dimensions of breakdown behaviour, including apparent prevalence at the initiating test, within-herd transmission, breakdown persistence, recurrence and herd size.

- **Severity at detection:** the proportion of animals testing positive at the index test (`prop_positive_index`), calculated as the ratio of total index reactors to the number of animals tested. This variable captures the intensity of infection at breakdown onset.
- **Within-herd transmission:** two complementary measures were used to characterise spread following detection. The `spread_ratio` was defined as the ratio of subsequent reactors to index reactors, quantifying the relative increase in detected infection after the initial test. The `spread_any` variable is a binary indicator taking the value 1 if any additional reactors were identified after the index test, and 0 otherwise.
- **Breakdown duration:** total duration of the breakdown in days (`bd_duration_days`), representing the persistence of infection.
- **Recent recurrence:** an indicator of whether a breakdown occurred within the previous 13 months (`bd_within_13_months`), capturing short-term recurrence risk.
- **Long-term recurrence:** the number of breakdowns initiated in the preceding five years (`no_bds_started_in_prev_5yr`), reflecting historical susceptibility.

- **Herd context:** mean herd size across multiple seasonal time points (`herd_size_mean`), representing herd scale and potential exposure.

Together, these variables capture complementary aspects of bTB breakdown dynamics, including severity, relative and absolute transmission dynamics, persistence, recurrence patterns, and herd-level structure.

Handling of Missing Data

No global imputation procedure was applied. Variables representing prior breakdown history were structurally interpreted, with absence of prior events corresponding to a value of zero.

Derived variables were constructed using conditional logic to avoid undefined values. In particular, proportions and ratios were set to missing where denominators were zero, and infinite values arising from division were treated as missing.

For continuous variables such as herd size, records with genuine missing values were excluded from clustering analyses to avoid introducing biologically implausible values through statistical imputation. This ensured that clustering structure was driven solely by observed data.

Data Transformation and Standardisation

To reduce skewness and stabilise variance in continuous variables, a $\log(1 + x)$ transformation was applied where appropriate.

All variables were subsequently standardised to have mean zero and unit variance to ensure equal contribution to distance-based clustering methods.

Exploratory data analysis was conducted to assess relationships between variables and identify potential redundancy prior to clustering. Pairwise correlations and scatter plots were examined. All figures are provided in the Supplementary Material. After initial data transformation and exploration and prior to the dimension reduction steps, a reproducible random sample of up to 20,000 observations was selected using a fixed seed to ensure computational efficiency while preserving the underlying structure of the data.

2.2. Principal Component Analysis

Principal Component Analysis (PCA) was applied to the set of seven standardised variables to reduce dimensionality and address potential multicollinearity. This dimensionality reduction step enabled efficient representation of the dominant patterns in the dataset while minimising noise and redundancy. The contribution of individual variables to each principal component was examined to aid interpretation of the underlying structure. This approach

provided a reduced feature space that preserves the key epidemiological variation in the data and facilitates subsequent analysis.

2.3. Clustering method

2.3.1. Partitioning Around Medoids (PAM)

Clustering was performed using the Partitioning Around Medoids (PAM) algorithm applied to the PCA-reduced feature space [20]. PAM was selected due to its robustness to outliers and its ability to identify representative observations (medoids) [21],[22], making it well-suited for heterogeneous epidemiological data.

To determine the optimal number of clusters, solutions for $K = 2$ to $K = 5$ were evaluated using silhouette width and visual inspection in PCA space (Figures 9 and 1). The optimal number for K was determined using a combination of the gap statistics, silhouette analysis, elbow method and NbClust. The elbow method was applied using both the traditional within-cluster sum of squares (WSS) and a normalised distance-based approach implemented in the ClusterR package (Figure 7 and Figure 8). Both methods assess cluster compactness as the number of clusters increases. The NbClust package was used to evaluate the optimal number of clusters based on multiple validity indices (Figure 18).

3. Multinomial Regression

This analysis investigated the epidemiological factors associated with membership of distinct bovine tuberculosis (bTB) breakdown cluster types, previously identified through Partitioning Around Medoids (PAM) clustering. Herds with no recorded breakdown history were assigned to a separate reference category (Cluster 0), serving as the control group, while herds that had experienced confirmed bTB breakdowns were grouped into three epidemiologically distinct clusters based on outbreak characteristics.

The primary objective was to determine which movement and neighbourhood exposure variables were significantly associated with an increased probability of belonging to each breakdown cluster relative to control herds. A multinomial logistic regression model was employed, extending binary logistic regression to accommodate a polytomous outcome by simultaneously estimating the log-odds of belonging to each infected cluster against the control reference .

Data Preparation for multinomial regression

The analytical dataset was constructed by combining two sources. The first comprised herds with confirmed bTB breakdowns assigned to one of three PAM-derived clusters (Clusters 1–3). The second comprised control herds those with no breakdown history over the

observation period assigned to Cluster 0. A random sample of 30,000 control observations was drawn to ensure computational tractability while preserving representativeness of the control population.

The resulting dataset contained 66,688 observations distributed across the four cluster categories (Table 1).

Table 1: Distribution of cluster membership in the analytical dataset.

Cluster	Description	n
0	Control (no breakdown history)	30,000
1	Breakdown cluster 1	23,370
2	Breakdown cluster 2	5,565
3	Breakdown cluster 3	7,753
Total		66,688

Within the breakdown herds, Cluster 1 comprised 16,128 New Breakdown and 7,242 Burnout herds; Cluster 2 comprised 3,805 New Breakdown and 1,760 Burnout herds; and Cluster 3 comprised 4,046 New Breakdown and 3,707 Burnout herds.

Predictor Variables

Seven predictor variables were included, capturing two broad domains of epidemiological exposure. The first domain comprised movement-related variables reflecting the infection status of the source herd for cattle received during the observation year: movements from Control (**Control_moves**), New Breakdown (**NewBD_moves**), Burnout (**Burnout_moves**), and Unknown (**Unknown_moves**) herds. The second domain captured local neighbourhood infection pressure, expressed as the number of neighbouring herds within a defined spatial radius classified as Burnout (**Burnout_neighbours**), Control (**Control_neighbours**), or New Breakdown (**NewBD_neighbours**) in the current year. Together, these predictors characterise both direct trade-related exposure and indirect spatial transmission risk.

Statistical Model

A multinomial logistic regression model was fitted with cluster membership as the polytomous outcome variable. Cluster 0 was specified as the reference category, so all estimated associations reflect the log-odds of belonging to an infected breakdown cluster relative to a herd with no recorded breakdown history. The model was estimated simultaneously across all outcome categories.

$$\log\left(\frac{P(\text{Cluster} = k)}{P(\text{Cluster} = 0)}\right) = \beta_{k0} + \beta_{k1} \text{Control_moves} + \beta_{k2} \text{NewBD_moves} \quad (1)$$

$$+ \beta_{k3} \text{Burnout_moves} + \beta_{k4} \text{Unknown_moves} \quad (2)$$

$$+ \beta_{k5} \text{Burnout_neighbours} + \beta_{k6} \text{Control_neighbours} \quad (3)$$

$$+ \beta_{k7} \text{NewBD_neighbours} \quad (4)$$

for $k \in \{1, 2, 3\}$. The model was implemented in R using the `multinom()` function from the `nnet` package, with a maximum of 500 iterations. Statistical significance was evaluated using Wald z -tests, and odds ratios (ORs) with 95% confidence intervals (CIs) were obtained by exponentiating the estimated coefficients and their standard errors.

Results

The model converged successfully after 30 iterations. The residual deviance was 143,514.6 (AIC = 143,562.6). The full model provided a significantly better fit than the null (intercept-only) model, as confirmed by the likelihood ratio test ($\chi^2 = 10,605$; $\text{df} = 21$; $p < 0.001$). McFadden's pseudo- R^2 was 0.069, indicating that the movement and neighbourhood predictors collectively discriminate between control herds and the three breakdown cluster types.

3.1. Odds Ratios and Confidence Intervals

Odds ratios, 95% CIs, and p -values for each predictor are presented in Table 2. All comparisons are relative to Cluster 0.

Table 2: Odds ratios (OR) and 95% confidence intervals from multinomial logistic regression predicting breakdown cluster membership relative to control herds (Cluster 0).

Predictor	Cluster 1 vs. 0			Cluster 2 vs. 0			Cluster 3 vs. 0		
	OR	95% CI	p	OR	95% CI	p	OR	95% CI	p
Intercept	0.532	(0.512, 0.552)	<0.001	0.095	(0.089, 0.101)	<0.001	0.142	(0.134, 0.150)	<0.001
Control_moves	1.000	(0.999, 1.001)	0.840	1.003	(1.002, 1.004)	<0.001	0.999	(0.998, 1.000)	0.222
NewBD_moves	1.007	(1.003, 1.011)	<0.001	1.009	(1.004, 1.014)	<0.001	1.010	(1.005, 1.015)	<0.001
Burnout_moves	1.004	(1.002, 1.007)	<0.001	1.006	(1.003, 1.009)	<0.001	1.009	(1.007, 1.012)	<0.001
Unknown_moves	1.000	(0.979, 1.022)	0.978	1.026	(1.003, 1.051)	0.030	0.993	(0.960, 1.028)	0.696
Burnout_neighbours	1.024	(1.021, 1.028)	<0.001	1.046	(1.041, 1.051)	<0.001	1.039	(1.034, 1.044)	<0.001
Control_neighbours	0.986	(0.985, 0.987)	<0.001	0.984	(0.982, 0.986)	<0.001	0.981	(0.979, 0.982)	<0.001
NewBD_neighbours	1.364	(1.354, 1.374)	<0.001	1.432	(1.419, 1.446)	<0.001	1.479	(1.466, 1.492)	<0.001

95% confidence intervals computed from Wald standard errors. Cluster 0 = control herds (reference category). p -values obtained from two-tailed Wald z -tests.

Movement Effects

Movements from New Breakdown herds (**NewBD_moves**) Figure 24 were positively associated with cluster membership across all three clusters (OR = 1.007, 1.009, and 1.010 for Clusters 1, 2, and 3, respectively; all $p < 0.001$). Similarly, movements from Burnout herds (**Burnout_moves**) showed consistent positive associations (OR = 1.004, 1.006, and 1.009; all $p < 0.001$). Although per-unit effects are modest, they represent marginal increases per individual animal movement and are likely to reflect substantially elevated risk at higher movement volumes.

Movements from Control herds (**Control_moves**) were not a meaningful predictor for Clusters 1 or 3 ($p = 0.840$ and $p = 0.222$), consistent with the expectation that sourcing cattle from uninfected herds confers lower transmission risk. A small but statistically significant positive association was observed for Cluster 2 (OR = 1.003; $p < 0.001$). Movements from Unknown-status herds (**Unknown_moves**) were significant only for Cluster 2 (OR = 1.026; $p = 0.030$) and non-significant for Clusters 1 and 3.

Neighbourhood Effects

Neighbourhood exposure variables demonstrated the strongest and most consistent associations with cluster membership Figure 25. The number of neighbouring New Breakdown herds (**NewBD_neighbours**) was the most influential predictor, with each additional infected neighbouring herd associated with a 36%, 43%, and 48% increase in the odds of belonging to Clusters 1, 2, and 3, respectively (all $p < 0.001$). This gradient across clusters underscores the importance of local spatial transmission in shaping outbreak risk.

Burnout neighbourhood exposure (**Burnout_neighbours**) was also positively associated with membership in all three clusters (OR = 1.024, 1.046, and 1.039; all $p < 0.001$), consistent with ongoing or resolving breakdowns constituting a source of residual environmental and contact-based transmission pressure. Conversely, a greater number of Control-status neighbouring herds (**Control_neighbours**) was inversely associated with cluster membership (OR = 0.986, 0.984, and 0.981; all $p < 0.001$), suggesting that an uninfected local neighbourhood is associated with reduced breakdown risk.

Predicted Probabilities

Average predicted probabilities across all observations were 0.443 for Cluster 0, 0.353 for Cluster 1, 0.085 for Cluster 2, and 0.118 for Cluster 3, closely reflecting the observed cluster proportions in the dataset. When **NewBD_neighbours** was varied across its observed range while all other predictors were held at their sample means, the predicted probability of control cluster membership declined and probabilities of infected cluster membership increased correspondingly, illustrating the dose-response nature of local infection pressure on

breakdown risk.

This analysis integrated PAM-derived outbreak phenotypes with multinomial logistic regression to characterise the epidemiological drivers of distinct bTB cluster types simultaneously. Using unsupervised cluster assignments as the outcome provides a more nuanced account of disease dynamics than binary case-control approaches, distinguishing herds with fundamentally different outbreak profiles rather than simply contrasting infected and uninfected herds.

Two mechanisms consistently emerged as principal drivers of cluster membership. First, movements from herds in active or resolving breakdown status were associated with increased odds across all clusters, consistent with evidence that trade networks constitute an important transmission pathway . Second, and more prominently, local neighbourhood infection pressure particularly from neighbouring New Breakdown herds — was strongly associated with elevated outbreak risk across all three cluster types. This aligns with the established roles of cattle-to-cattle boundary contact and wildlife reservoir pressure in sustaining local bTB transmission.

The inverse association for Control-status neighbouring herds may reflect genuine protective effects of an uninfected local environment, or alternatively may indicate that herds in low-prevalence areas are less likely to experience the outbreak intensity that characterises Clusters 1–3. Disentangling these explanations would require longitudinal data linking neighbourhood composition to subsequent breakdown incidence.

The per-unit odds ratios for movement variables, while statistically significant, are modest in magnitude. This is consistent with the incremental nature of individual animal movements as a risk factor and does not preclude substantially elevated risk at higher movement volumes. Future analyses could model movement exposure as a categorised or polynomial term to better capture nonlinear dose-response relationships.

Multinomial logistic regression identified movement and neighbourhood risk factors associated with distinct bTB outbreak cluster types. Neighbourhood exposure to New Breakdown herds was the strongest driver of infected cluster membership, followed by Burnout neighbourhood exposure and movements from infected herds. Control-status neighbourhood density demonstrated a consistent protective association across all three cluster comparisons. These findings support the integration of spatial infection pressure and movement network data into bTB surveillance frameworks, and the identified exposure profiles may inform more targeted herd-level risk stratification and proportionate disease control interventions.

3.1.1. Survival Analysis of Time to Next Breakdown by Cluster

Kaplan–Meier survival curves were used to compare the time to subsequent breakdown across the three clusters identified by PAM clustering (Figure 26). The survival function represents the probability of a herd remaining breakdown-free over time.

3.1.2. Software and reproducibility

All analyses were conducted using R (version 4.5.0) within RStudio on a Windows environment, with a fixed random seed (`set.seed(123)`) to ensure reproducibility. The analysis utilised several packages, including `ClusterR` (for K-means optimisation), `factoextra` (for visualisation and cluster validation), `cluster` (for PAM clustering), `mclust` (for Gaussian Mixture Models), `dplyr` (for data manipulation), and `ggplot2` (for visualisation) [23],[24]. All code used for data processing, analysis, and visualisation is publicly available at:

<https://github.com/victorychukwudi/Modeling-Cattle-TB-Spread-with-Networks>.

4. Results

Descriptive Statistics

Descriptive statistics of the clustering variables are presented in Table 3. Overall, substantial variability was observed across herds. Breakdown duration and spread ratio exhibited wide ranges, indicating heterogeneity in outbreak severity. In contrast, binary indicators such as within-herd spread and recurrence within 13 months were relatively infrequent. Herd size also varied considerably, supporting the need for clustering to identify distinct outbreak profiles.

Table 3: Descriptive statistics of clustering variables

Variable	Mean	SD	Min	Max
Proportion positive at index test	0.07	0.11	0.00	1.00
Spread ratio	0.34	1.81	0.00	102
Any within-herd spread (0/1)	0.18	0.39	0.00	1.00
Breakdown duration (days)	184	101	60	3042
Breakdown within 13 months (0/1)	0.04	0.19	0.00	1.00
Previous breakdowns (5 years)	0.22	0.47	0.00	6
Mean herd size	101	110	0	2231

4.1. Exploratory Data Analysis

Initial exploratory analysis revealed that several variables were strongly right-skewed, with most herds exhibiting low-severity breakdowns and a small number of high-severity

outliers (Figure 2). Log-transformation substantially reduced skewness and stabilised variance across continuous variables, improving their suitability for distance-based clustering while preserving key epidemiological patterns (Figure 3).

Density plots of the transformed variables further confirmed improved distributional symmetry and reduced extreme dispersion (Figure 4). Examination of pairwise relationships using scatter plot matrices indicated varying degrees of association between variables, with some evidence of clustering structure and non-linear relationships (Figure 5).

Correlation analysis demonstrated moderate associations among selected variables, with no evidence of excessive multicollinearity that would compromise clustering performance (Figure 6). Collectively, these exploratory steps supported the suitability of the transformed variables for subsequent multivariate and clustering analyses.

4.2. PCA Results

The distribution of variables in the PCA input dataset (Figure 10) indicates that log-transformation and scaling reduced extreme skewness while preserving meaningful variability across features. Continuous variables exhibit approximately symmetric or moderately right-skewed distributions, whereas binary and count-based variables retain discrete structures. This confirms that preprocessing improved comparability without obscuring underlying epidemiological heterogeneity.

The scree plot (Figure 11) shows that the first four principal components explain a substantial proportion of total variance (34.3%, 22.5%, 17.7%, and 9.2%, respectively), yielding a cumulative variance of approximately 83.6%. Beyond the fourth component, individual contributions drop below 10%, indicating diminishing returns. This supports the retention of four components as a suitable representation of the data.

Visualisation of observations in the space defined by the first two principal components (Figure 13) reveals a structured yet continuous distribution. Rather than forming clearly separated groups, observations are dispersed along gradients, with pronounced elongation along the first principal component (PC1).

A dense concentration of observations is observed near the origin, alongside a broader spread extending primarily along PC1. This suggests that the dominant variation is captured along a single gradient, while secondary variation contributes to additional dispersion without forming distinct boundaries. The presence of more extreme observations further indicates heterogeneity in breakdown characteristics.

Table 4: Variance explained by principal components

	PC1	PC2	PC3	PC4	PC5	PC6	PC7
Standard deviation	1.548	1.254	1.114	0.802	0.750	0.592	0.484
Proportion of variance	0.343	0.225	0.177	0.092	0.080	0.050	0.033
Cumulative proportion	0.343	0.567	0.744	0.836	0.917	0.967	1.000

Table 5: Principal component loadings

Variable	PC1	PC2	PC3	PC4	PC5	PC6	PC7
prop_positive_index	0.040	0.565	0.475	0.151	-0.484	-0.433	-0.094
spread_ratio	-0.553	0.133	-0.091	-0.077	0.419	-0.341	-0.609
spread_any	-0.575	0.166	-0.062	-0.005	0.118	-0.204	0.763
bd_duration_days	-0.539	0.118	0.070	0.060	-0.410	0.698	-0.176
bd_within_13_months	-0.113	-0.350	0.608	-0.702	-0.018	-0.032	0.030
no_bds_started_in_prev_5yr	-0.123	-0.421	0.533	0.686	0.231	0.002	0.011
herd_size_mean	-0.208	-0.567	-0.322	0.068	-0.596	-0.407	-0.076

4.2.1. Structure of the PCA space

The distribution of observations in the reduced PCA space (Figure 12) is characterised by a dense central region and a pronounced spread along the first principal component. This indicates that the primary variation in the dataset is captured along PC1, while variation along PC2 is comparatively more limited.

The observed structure reflects a continuous gradient rather than clearly separated groups, with elongation along PC1 suggesting systematic variation in breakdown characteristics. Secondary dispersion along PC2 and higher components contributes additional heterogeneity, but without forming distinct clusters in the raw feature space.

This geometry supports the application of clustering methods in the reduced PCA space, where the dominant structure is preserved while noise and collinearity are reduced.

4.2.2. Variable contributions to principal components

To aid interpretation of the principal components, the contribution of each variable to the first three dimensions was examined (Figure 15).

For PC1, the dominant contributors were `spread_any`, `spread_ratio`, and `bd_duration_days`, all of which exceed the expected average contribution. This indicates that the first principal component primarily captures variation associated with the occurrence, magnitude, and duration of breakdown spread. In contrast, variables such as `prop_positive_index`,

`bd_within_13_months`, and `no_bds_started_in_prev_5yr` contribute relatively little, suggesting a more limited role in defining this dominant axis.

For PC2, the largest contributions arise from `herd_size_mean` and `prop_positive_index`, both substantially exceeding the average contribution threshold. This component therefore reflects variation in herd-level characteristics, particularly herd size and infection intensity. A moderate contribution from `no_bds_started_in_prev_5yr` suggests an additional influence of historical breakdown frequency, although to a lesser extent than the primary variables.

For PC3, the dominant contributors are `bd_within_13_months`, `no_bds_started_in_prev_5yr`, and `prop_positive_index`. This indicates that the third component captures temporal and recurrence-related dynamics, distinguishing herds based on recent breakdown history and persistence over time. Contributions from variables associated with spread magnitude are minimal in this dimension, highlighting its distinct interpretation relative to PC1.

These relationships are consistent with the PCA biplot (Figure 16), where variables related to spread dynamics align primarily with PC1 (Figure 15), herd-level characteristics align with PC2, and temporal recurrence variables contribute more strongly to PC3. The separation of these vectors across components indicates that breakdown intensity, herd structure, and temporal recurrence represent largely independent dimensions of variation in the dataset.

Taken together, the PCA decomposes the data into three interpretable gradients: (i) a dominant axis associated with breakdown spread and duration (PC1), (ii) a secondary axis reflecting herd size and infection intensity (PC2), and (iii) a tertiary axis capturing temporal recurrence and breakdown history (PC3). This structure provides a coherent basis for subsequent clustering, allowing groups to be interpreted in terms of these underlying epidemiological processes.

The absence of sharp dominance beyond these components supports a reduced representation while retaining the key dimensions of variation, reinforcing the suitability of PCA as a preprocessing step for clustering in this context.

Clustering Results

The results indicated that $K = 3$ was the most frequently recommended solution across the evaluated indices, receiving the highest level of support. While alternative values such as $K = 4$ and $K = 5$ are suggested by a smaller number of indices, their support is notably weaker.

This consensus across multiple independent criteria provides strong evidence that a three-cluster solution best captures the underlying structure of the data.

The number of clusters was determined using the average silhouette width obtained from

partitioning around medoids (PAM) (Table 8). The silhouette values indicate moderate clustering structure across the range of tested solutions.

Although the highest silhouette width was observed at $K = 2$, the $K = 3$ solution was selected for subsequent analysis. This choice reflects the ability of the three-cluster configuration to capture additional structure in the data while maintaining acceptable cluster separation.

For $K \geq 4$, the silhouette width decreases, indicating reduced cohesion and increased overlap between clusters. This suggests that additional clusters primarily partition existing structure rather than revealing substantively distinct groupings.

Accordingly, $K = 3$ was retained as the optimal number of clusters for the PAM analysis, providing a parsimonious representation of the underlying heterogeneity in the dataset.

4.2.3. Description of Clusters

Table 6: Mean values of variables by cluster (PAM, $K = 3$)

Cluster	prop_pos	spread_ratio	spread_any	duration	within_13m	prev_5yr	herd_size
1	-0.24	-0.35	-0.47	-0.36	-0.19	-0.49	-0.06
2	-0.33	-0.35	-0.47	-0.21	-0.19	1.81	0.32
3	-0.07	1.38	2.12	1.16	-0.19	-0.49	0.39

The PAM clustering reveals a similar overall structure to K-means. Cluster 1 represents low values across most variables, indicating low transmission. Cluster 3 shows high spread and longer duration, consistent with high-risk impacts outbreaks. Cluster 2 exhibit a higher number of previous breakdowns despite generally low spread indicators, suggesting that PAM captures variation related to historical recurrence more distinctly.

4.2.4. Mean values of variables by cluster (K-means, $K = 3$)

To interpret the clustering results, the mean value of each variable was calculated within each cluster. This allows for comparison of cluster profiles and identification of distinguishing characteristics.

The results (Table 7) show clear differences between clusters across the variables.

Table 7: Mean values of variables by cluster

Cluster	prop_pos	spread_ratio	spread_any	duration	within_13m	prev_5yr	herd_size
1	-0.12	0.19	0.13	0.51	5.15	1.79	0.53
2	-0.01	-0.35	-0.47	-0.34	-0.19	-0.08	-0.09
3	0.09	1.54	2.10	1.38	-0.19	-0.02	0.28

Cluster 1 is characterised by moderate increases in outbreak duration, within-herd spread, and number of breakdowns, suggesting a group with relatively higher transmission intensity.

Cluster 2 shows consistently negative values across most variables, indicating lower-than-average levels for spread, duration, and outbreak frequency. This cluster represents relatively low-risk or less affected observations.

Cluster 3 displays the highest values for spread-related variables and outbreak duration, indicating more extensive transmission and prolonged outbreaks. This cluster may represent high-risk or more severe cases.

Overall, the clusters capture meaningful differences in disease dynamics, separating low, moderate, and high transmission profiles.

4.2.5. Silhouette comparison of K-means and PAM

The average silhouette width was used to compare clustering performance between K-means and PAM across different values of K (Table 8).

Table 8: Average silhouette width for K-means and PAM

Method	K	Avg. Silhouette
K-means	2	0.460
K-means	3	0.490
K-means	4	0.338
K-means	5	0.441
PAM	2	0.457
PAM	3	0.417
PAM	4	0.299
PAM	5	0.347

K-means consistently achieves higher silhouette scores than PAM, particularly at $K = 3$, where it reaches its maximum value. This indicates better-defined and more cohesive clusters for K-means.

Although PAM produces a broadly similar clustering structure, its lower silhouette values suggest slightly weaker separation between clusters. These results support the selection of K-means with $K = 3$ as the preferred clustering approach.

4.2.6. Description of Cluster Results

The three-cluster solution reveals distinct epidemiological breakdown profiles based on the underlying variables.

Cluster 1 is characterised by consistently negative values across all variables, including `spread_ratio`, `spread_any`, `bd_duration_days`, and `no_bds_started_in_prev_5yr`. This indicates herds with low infection intensity, limited spatial spread, short breakdown duration, and little evidence of recurrence, representing a low-risk group.

Cluster 2 displays a distinct profile with low spread-related values but a markedly elevated value for `no_bds_started_in_prev_5yr`. This indicates herds experiencing frequent recurrent breakdowns over time, despite limited within-breakdown transmission. This pattern suggests infection driven by repeated re-introduction or persistent underlying risk factors rather than active spread.

Cluster 3 exhibits strongly positive values for `spread_ratio`, `spread_any`, and `bd_duration_days`, reflecting substantial within-herd transmission and prolonged outbreak duration. In contrast, recurrence remains low in this group, indicating that these are intense but relatively isolated breakdown events.

Together, these clusters highlight that breakdown dynamics are driven by distinct processes, namely low-risk absence of infection, recurrence over time, and high-intensity transmission.

4.3. Temporal distribution of clusters

The temporal distribution of clusters was examined by calculating the annual proportion of herds assigned to each PAM cluster ($K = 3$). The results are summarised in (Table 9) and illustrated in (Figure 23).

Table 9: Annual proportion of herds assigned to each cluster.

Year	Cluster 1	Cluster 2	Cluster 3
2007	0.58	0.15	0.26
2008	0.66	0.10	0.23
2009	0.64	0.16	0.19
2010	0.61	0.20	0.19
2011	0.62	0.18	0.20
2012	0.62	0.19	0.20
2013	0.65	0.17	0.18
2014	0.68	0.15	0.17
2015	0.68	0.17	0.15
2016	0.68	0.17	0.16
2017	0.68	0.14	0.18
2018	0.69	0.15	0.16
2019	0.67	0.15	0.18
2020	0.66	0.16	0.19
2021	0.67	0.15	0.17
2022	0.64	0.17	0.19
2023	0.64	0.18	0.19
2024	0.75	0.17	0.09

Cluster 1 consistently represents the majority of herds across all years, with proportions generally ranging between approximately 0.61 and 0.69, indicating a stable dominant profile over time. Clusters 2 and 3 remain smaller and relatively stable, each typically accounting for between 0.14 and 0.20 of observations.

While minor year-to-year fluctuations are observed, no substantial structural shifts in cluster composition are evident. A notable deviation occurs in 2024, where Cluster 1 increases markedly (approximately 0.75), accompanied by a reduction in Cluster 3. This pattern should be interpreted cautiously, as it may reflect incomplete or partially observed data for the most recent year.

Collectively, the stability of cluster proportions suggests that the clustering captures persistent underlying epidemiological structure rather than short-term variation.

Clear differences in survival patterns are observed across clusters (Figure 26). Cluster 1 shows the slowest decline in survival probability, indicating that herds remain breakdown-free for longer periods and experience delayed recurrence. In contrast, Cluster 2 exhibits the most rapid decrease, suggesting a higher likelihood of earlier subsequent breakdowns.

Cluster 3 displays an intermediate pattern, lying between the two extremes.

The relatively gradual decline of the curves reflects that many herds do not experience a subsequent breakdown within the observed period, indicating delayed or infrequent recurrence. The log-rank test demonstrates a statistically significant difference between clusters ($p < 0.0001$), confirming that time to subsequent breakdown varies meaningfully across the identified groups. These results provide further evidence of heterogeneity in recurrence dynamics and support the epidemiological relevance of the clustering structure.

4.4. Summary of results

4.4.1. PAM cluster profiles

To assess the robustness of the clustering structure, the mean value of each variable was calculated for clusters obtained using PAM with $K = 3$.

4.5. Comparison with other clustering methods

- **K-means clustering:** Visual inspection of the clustering results (Figure 20) shows that K-means produces well-separated groups, particularly at $K = 3$. The clusters are compact and approximately elliptical, reflecting the method’s assumption of spherical structure in the data. As K increases beyond 3, clusters become more fragmented and overlap increases, suggesting over-partitioning. These observations, together with the higher silhouette score at $K = 3$, support the selection of K-means with three clusters as the most appropriate solution.
- **Gaussian Mixture Model (GMM):** Model-based clustering using GMM identified an optimal solution with $K = 2$ components based on the Bayesian Information Criterion (BIC) (Figure 21). The selected model (VEI) assumes diagonal covariance with equal shape, allowing clusters to vary in volume while maintaining axis-aligned structure.

The resulting classification (Figure 22) indicates two broad groups, with cluster sizes of 13,093 and 6,907 observations. Compared to K-means and PAM, this solution represents a higher-level partitioning of the data, capturing the dominant underlying structure rather than finer subgroups.

As a probabilistic approach, GMM also quantifies uncertainty in cluster membership [25],[26], highlighting regions of overlap where observations cannot be clearly assigned to a single group. While this provides a more flexible and realistic representation of the data, the two-cluster solution was considered overly coarse for the study objectives. Therefore, the three-cluster solution obtained from partitioning methods was retained for interpretation.

The WSS curve (Figure 7) shows a sharp reduction in within-cluster variation between $K = 1$ and $K = 3$, indicating that the majority of the data structure is captured within this range.

Beyond $K = 3$, the rate of decrease in WSS becomes more gradual, suggesting diminishing returns from adding additional clusters. Although there are minor fluctuations in the curve at higher values of K (e.g., around $K = 6$ and $K = 8$), the overall trend remains relatively flat compared to the initial steep decline. This indicates that further partitioning provides limited improvement in cluster compactness.

The absence of a perfectly sharp “elbow” suggests that the data may not have strongly distinct cluster boundaries; however, the most notable inflection occurs around $K = 3$ to $K = 4$.

The normalised distance-based curve exhibits a similar pattern, with a marked improvement in compactness up to approximately $K = 3$ – 4 (Figure 8), after which gains become progressively smaller. Although the values are scaled, the overall trend is consistent with the WSS results.

Together, these findings indicate that a small number of clusters adequately captures the primary structure of the data, with $K = 3$ representing a reasonable balance between model simplicity and explanatory power.

The silhouette plot for $K = 3$, indicates a clear and meaningful clustering structure, with the average silhouette width (Figure 9) supporting an adequate level of separation between clusters. A substantial proportion of observations exhibit positive silhouette values, suggesting that they are appropriately assigned to their respective clusters.

While some overlap between clusters is present, this is expected in complex epidemiological data and does not detract from the overall structure captured. The clustering solution demonstrates a coherent partitioning of the data into distinct groups.

Increasing the number of clusters may lead to minor improvements in separation; however, such gains are limited and may reduce interpretability. In this context, $K = 3$ provides a well-balanced solution that captures the dominant structure in the data while maintaining clarity and interpretability.

This finding is consistent with the elbow method and supports the selection of a three-cluster solution for subsequent analysis.

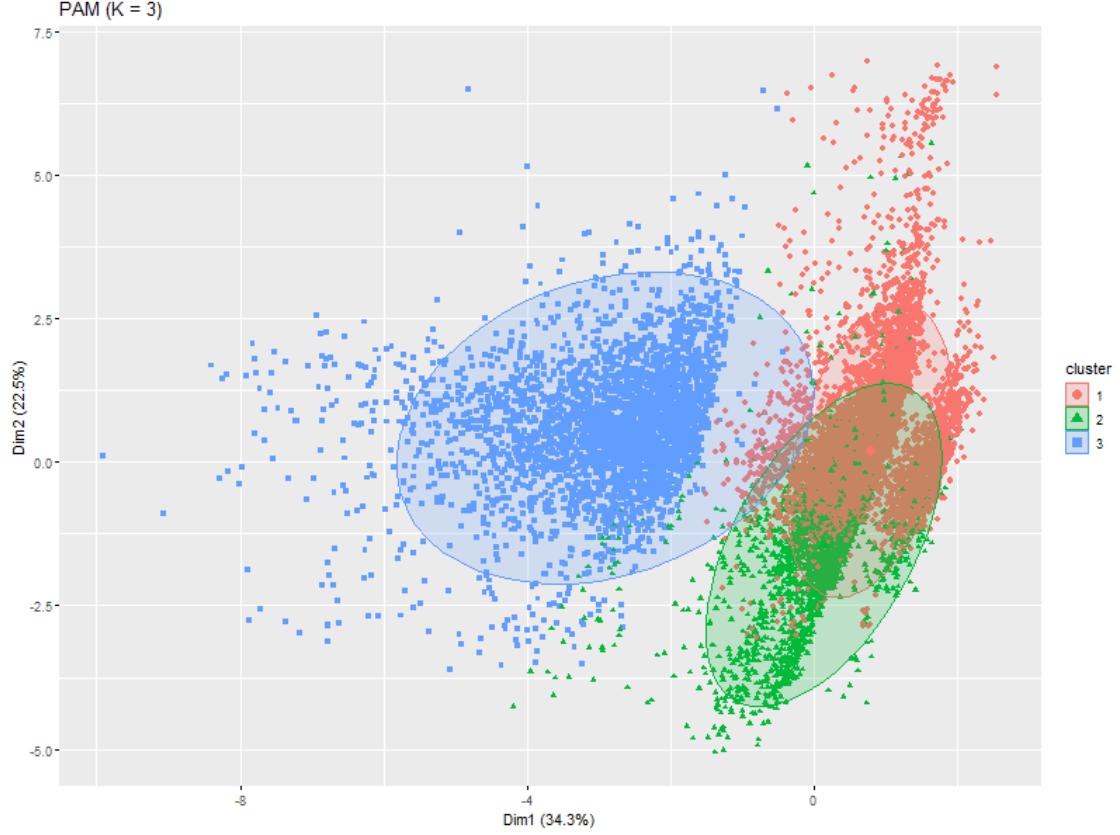


Figure 1: PAM clustering results for $K = 3$ projected onto the first two principal components.

5. Discussion

PAM clustering was selected over alternative methods such as k-means (Figure 20) and hierarchical clustering. Unlike k-means, which is sensitive to outliers and relies on cluster centroids, PAM identifies representative observations (medoids) (Figure 1), making it more robust for heterogeneous epidemiological data.

Compared to hierarchical clustering, PAM provides more stable and interpretable partitions when the number of clusters is predefined. These properties make PAM particularly suitable for the present dataset.

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Supplementary Material

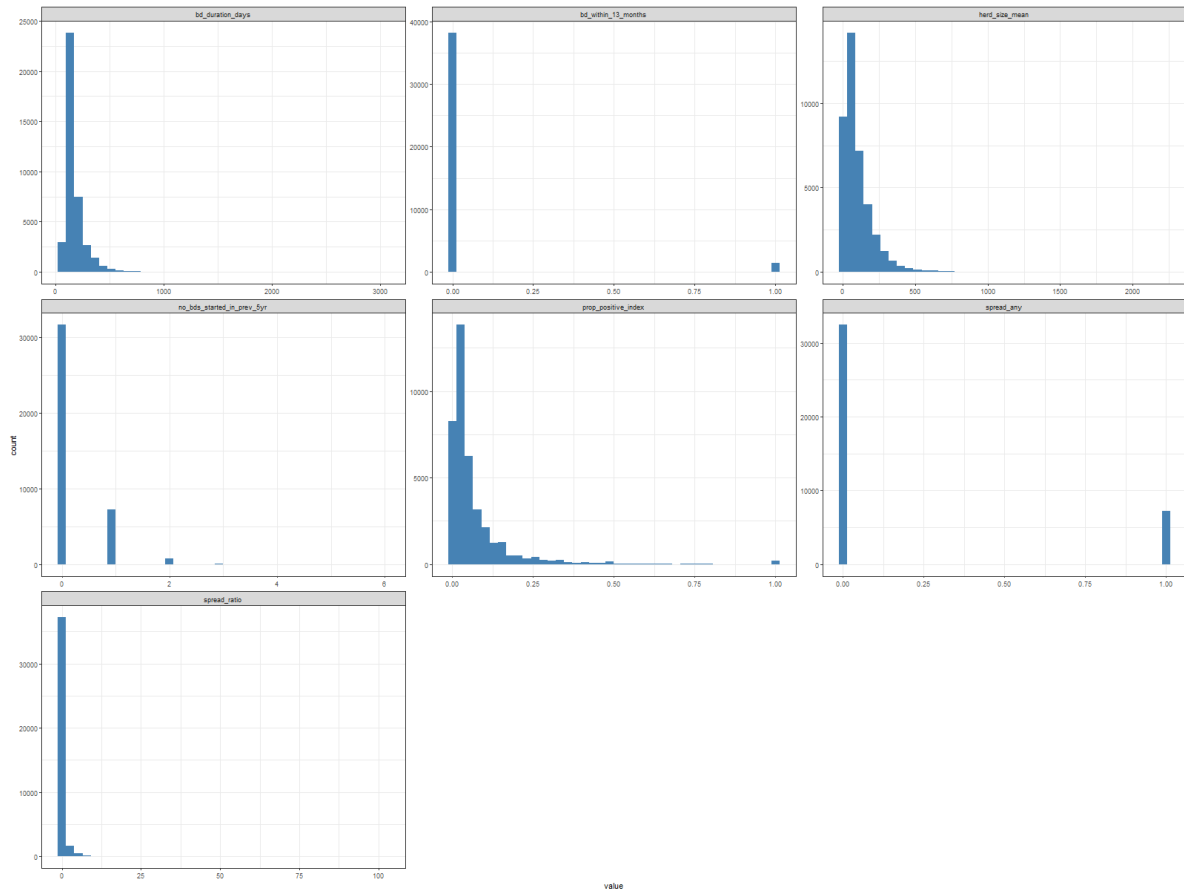


Figure 2: Histograms of raw variables prior to transformation, showing right-skewness and zero inflation in several variables.

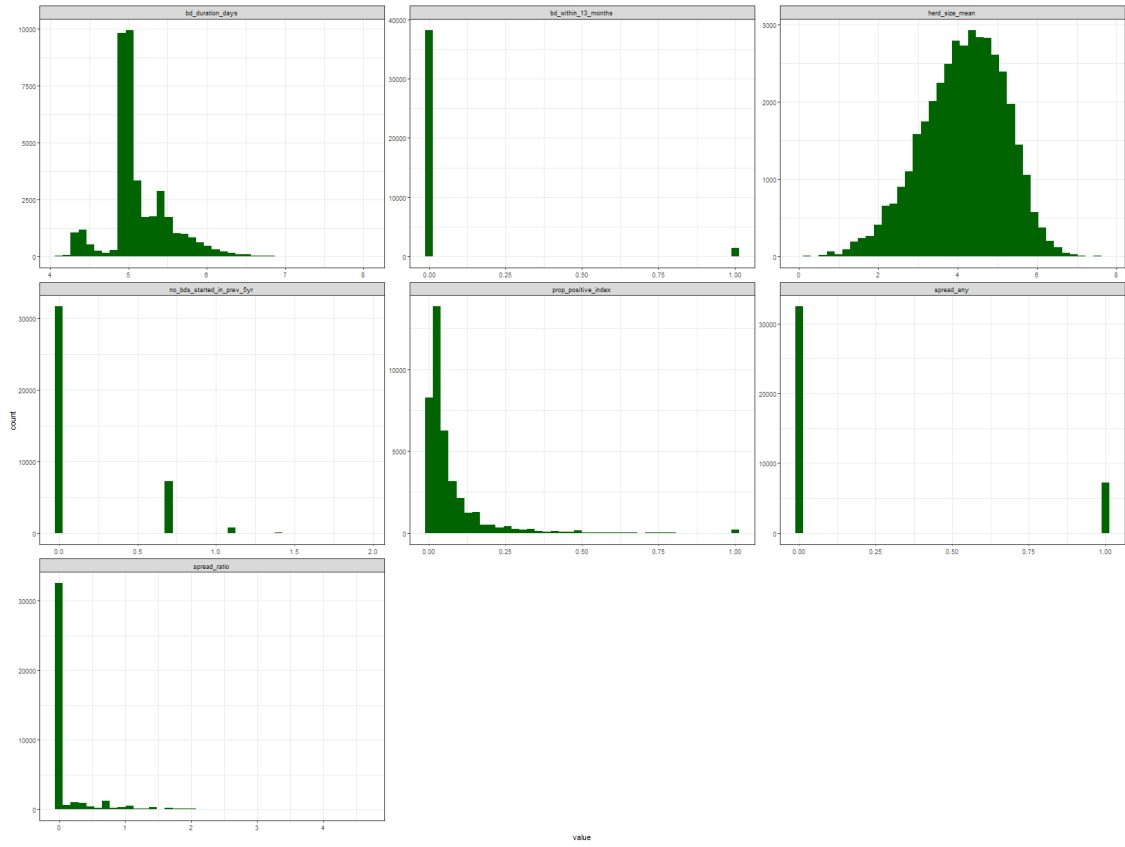


Figure 3: Histograms of variables after log-transformation.

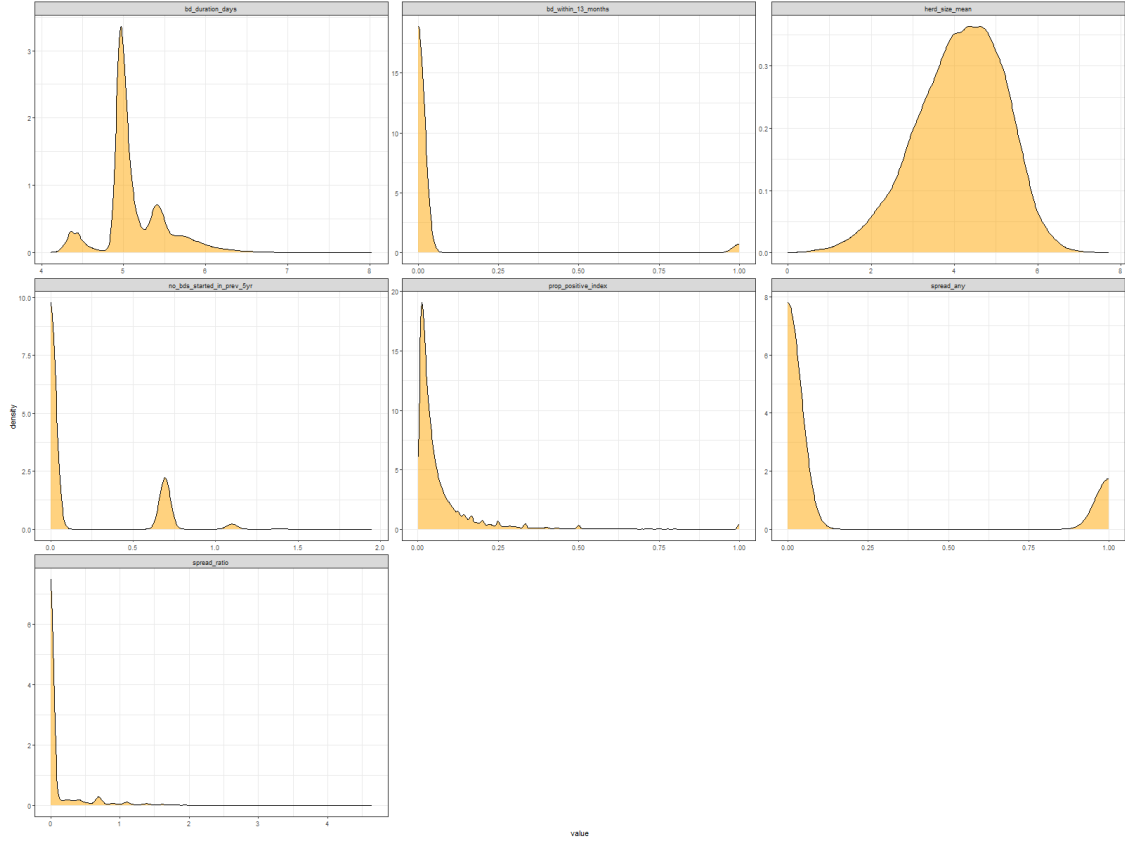


Figure 4: Density plots of transformed variables.

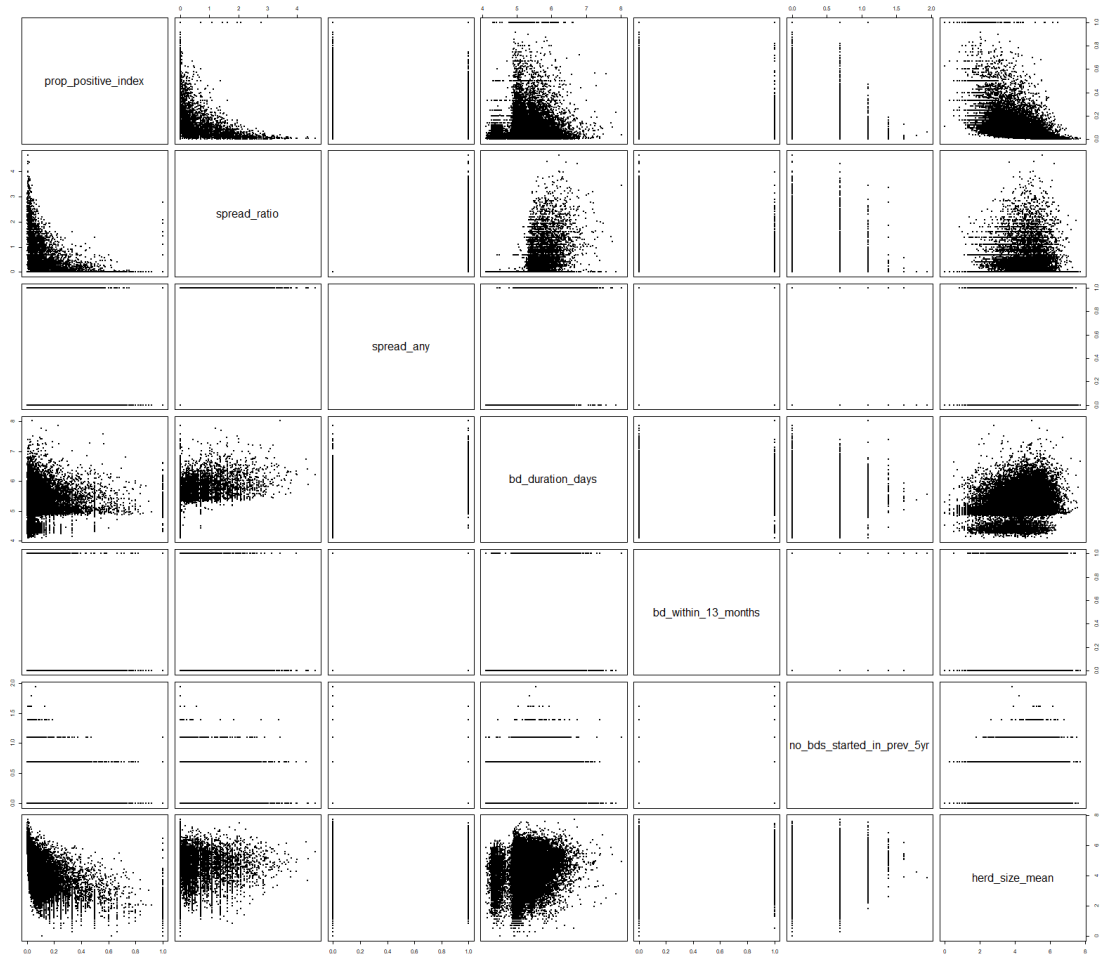


Figure 5: Scatter plot matrix illustrating relationships between variables.

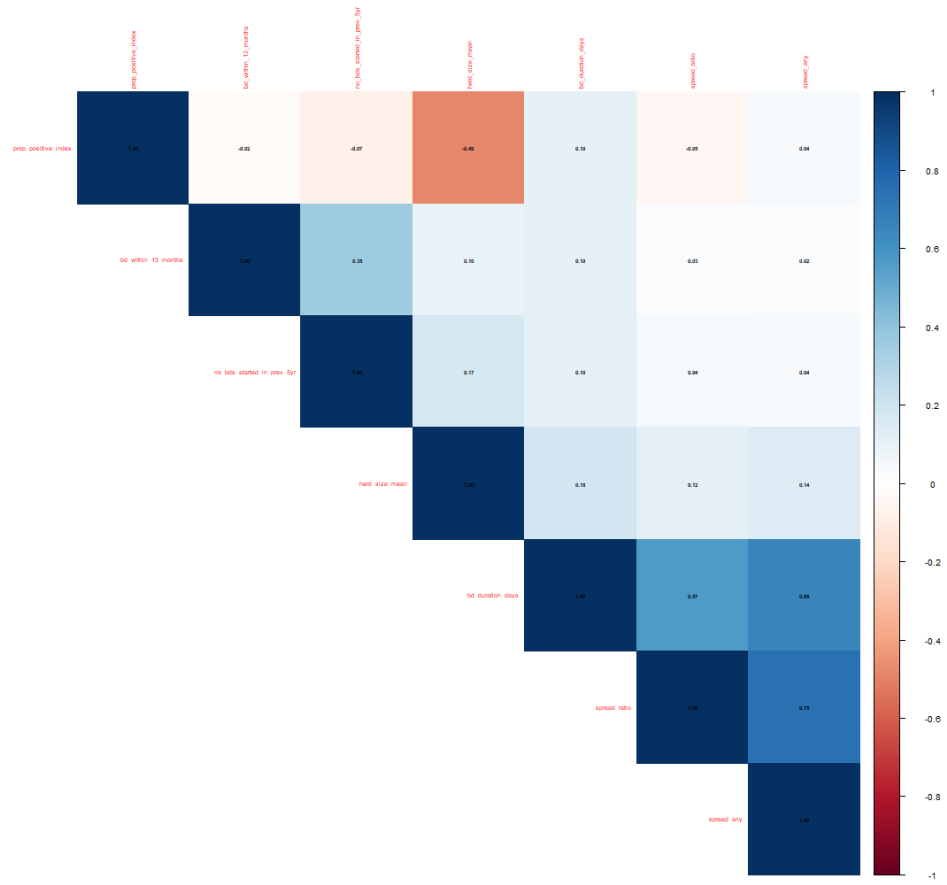


Figure 6: Correlation matrix of variables used in the clustering analysis.

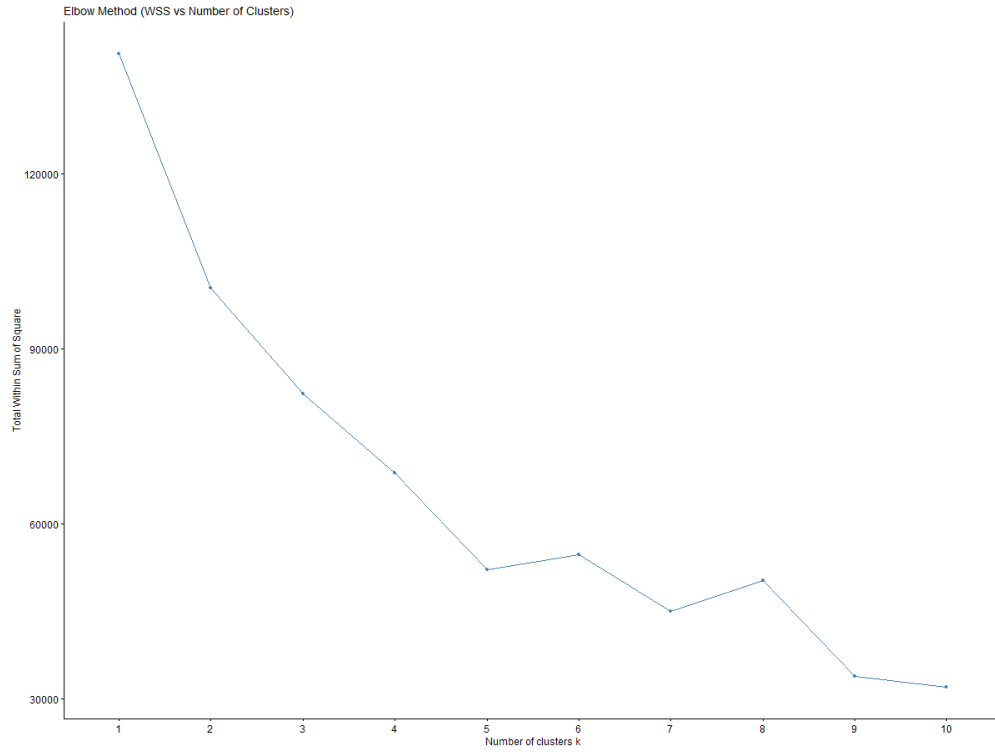


Figure 7: Elbow method based on within-cluster sum of squares (WSS) for $K = 1$ to 10. The curve shows a steep decline up to $K = 3$, followed by a plateau, indicating diminishing improvements in cluster compactness beyond this point.

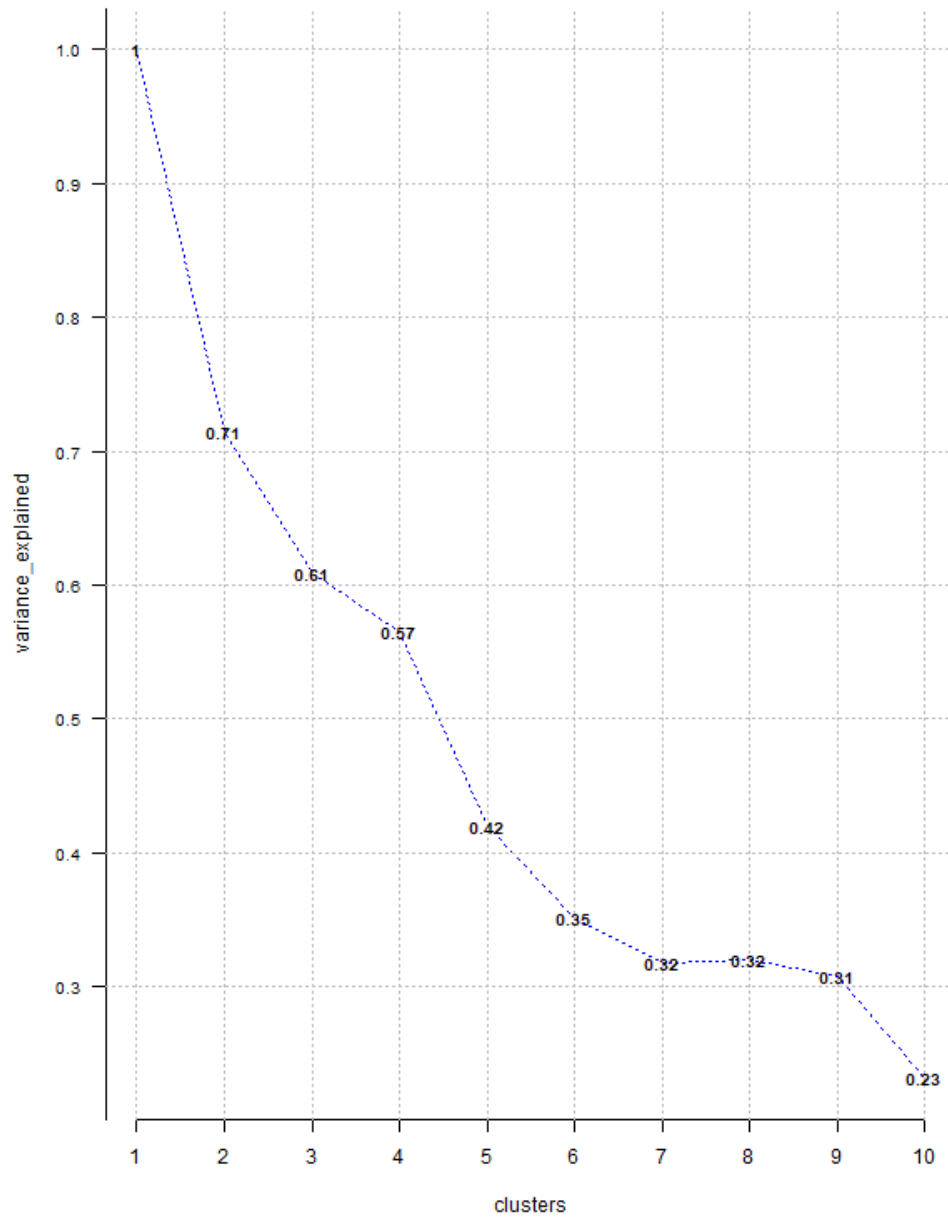


Figure 8: Normalised cluster compactness from the `ClusterR` elbow method.

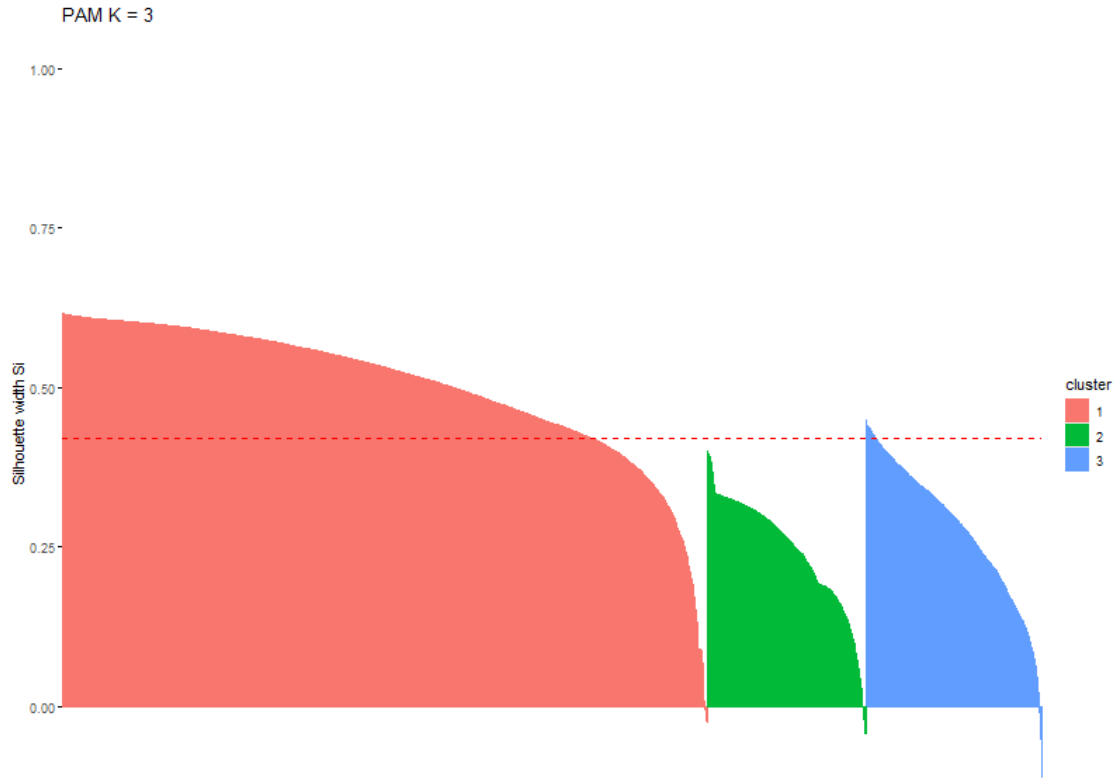


Figure 9: Silhouette plots for PAM clustering with $K = 3$. The dashed line indicates the average silhouette width.

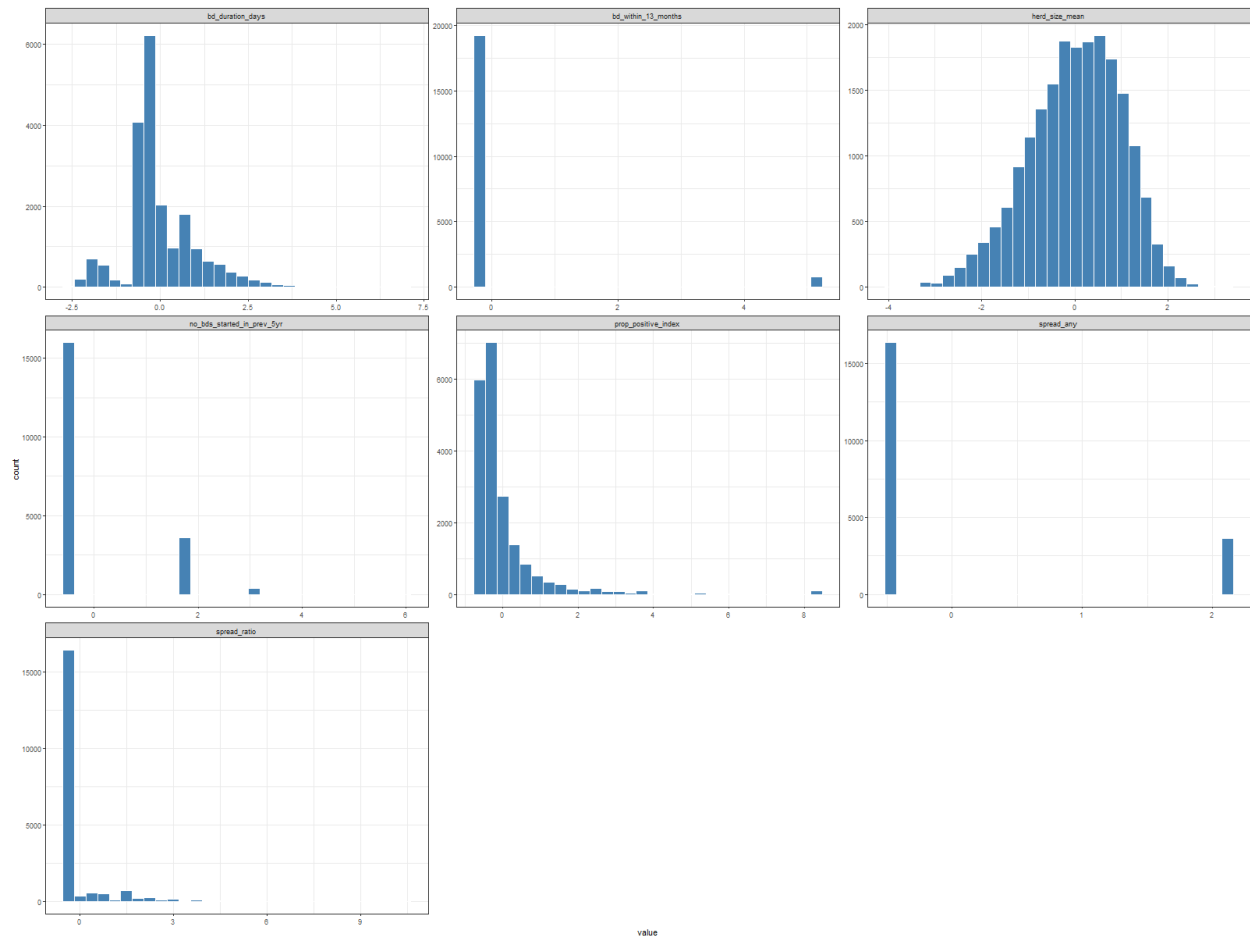


Figure 10: Distribution of variables used in PCA after transformation and scaling.

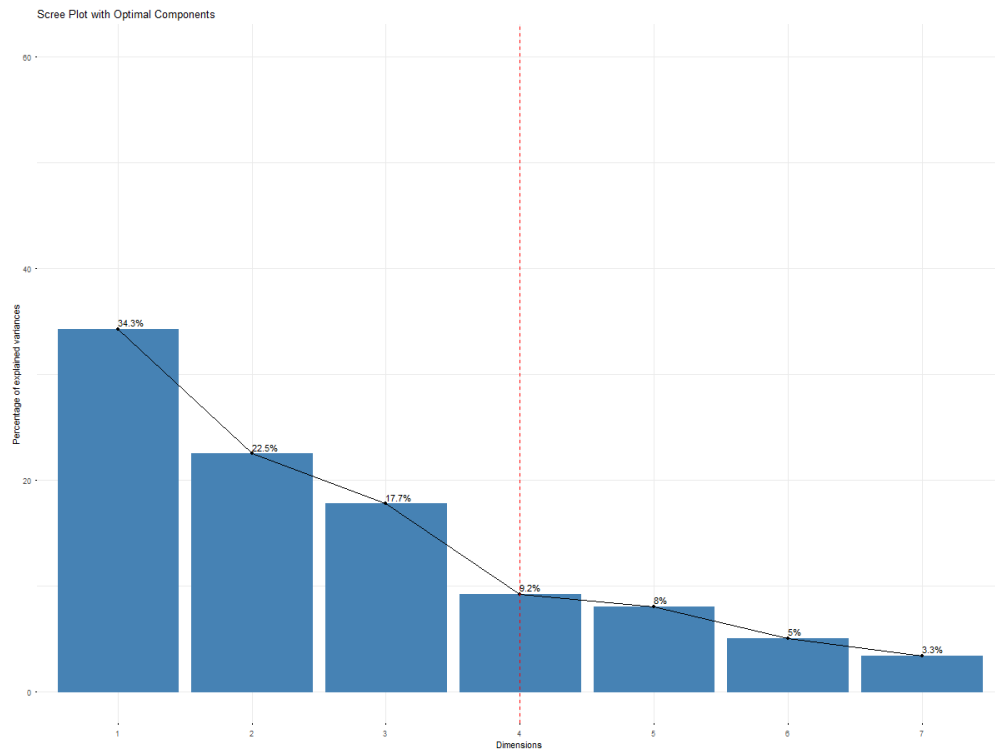


Figure 11: Scree plot showing the percentage of variance explained by each principal component.

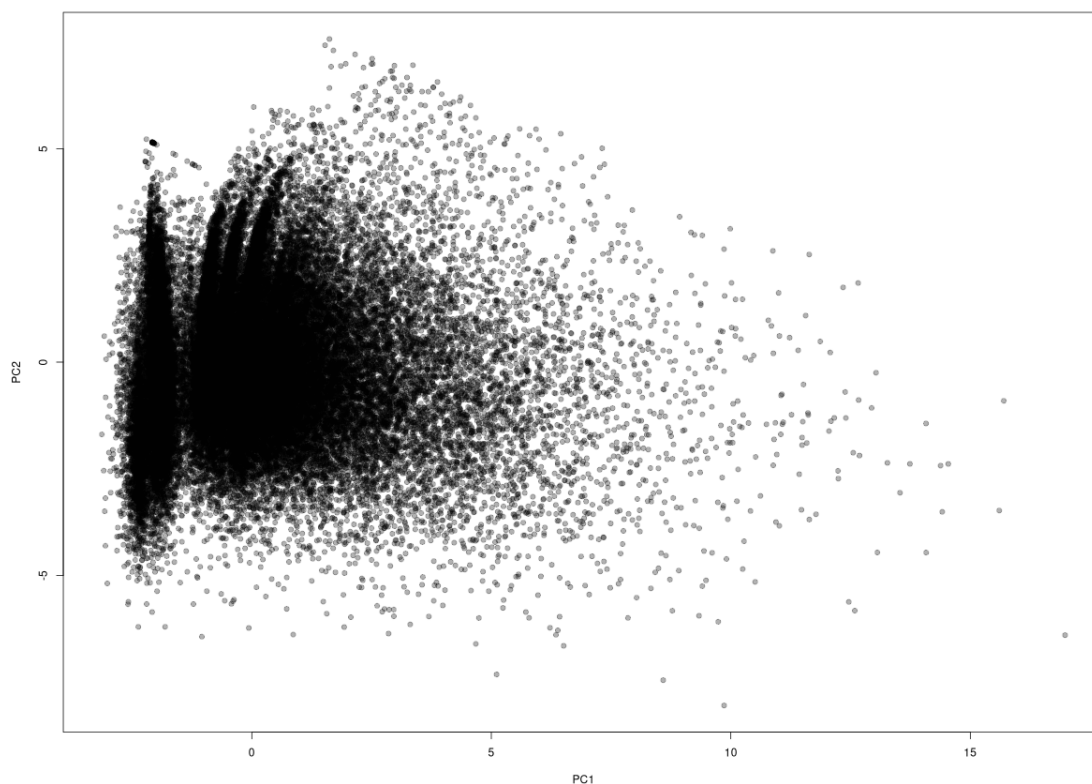


Figure 12: Distribution of observations projected onto the first two principal components.

Figure 12 illustrates the overall structure of the data in PCA space. The point cloud exhibits a highly concentrated central region with increasing dispersion along the positive direction of PC1, suggesting the presence of a dominant axis of variation. The asymmetric spread and presence of outlying observations indicate heterogeneity in the data, with a small number of observations extending beyond the main distribution.

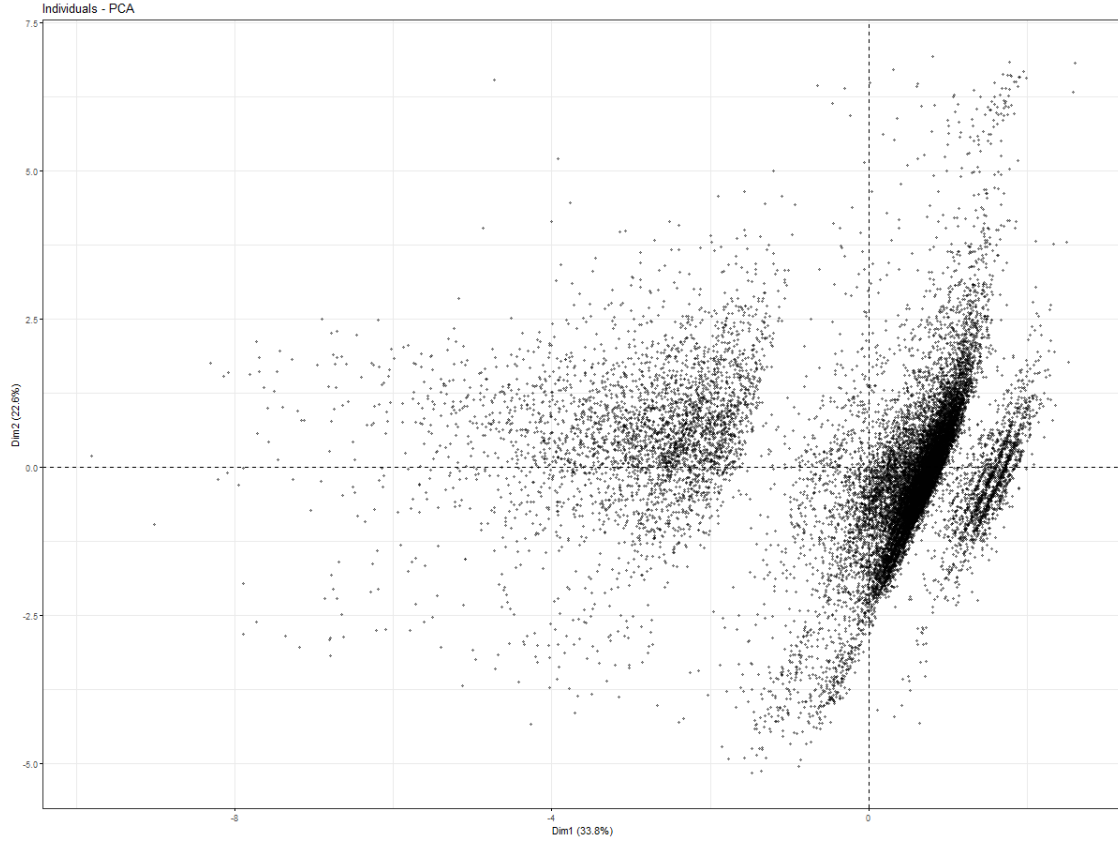


Figure 13: Projection of observations onto the first two principal components.

Figure 13 shows the projection of observations onto the first two principal components. The distribution reveals a dominant gradient along PC1, with a dense concentration of observations near the origin and a gradual spread towards higher values. Variation along PC2 is comparatively more limited, indicating that the primary structure in the data is largely driven by the first principal component, with additional but less pronounced heterogeneity captured by the second component.

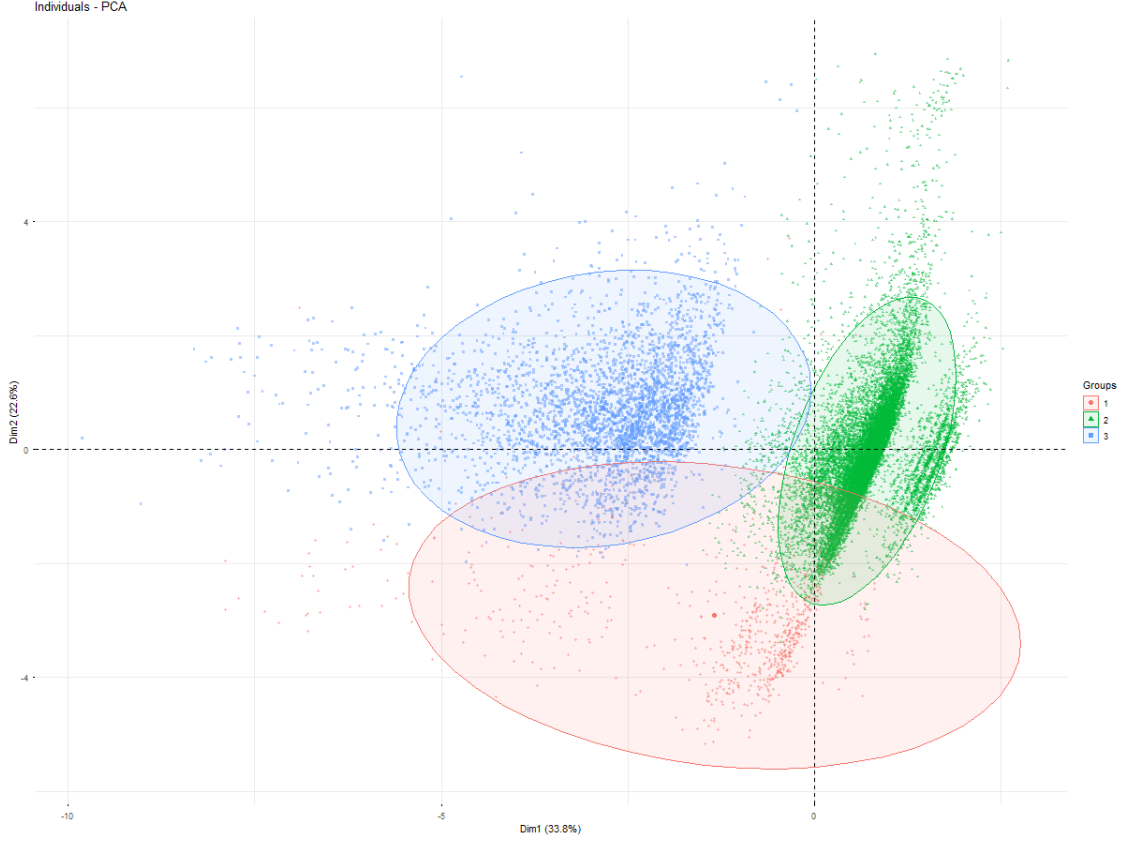


Figure 14: Clustering results for $K = 3$ projected onto the first two principal components.

The PCA plot indicates a moderate degree of separation between the three clusters in the reduced two-dimensional space. One cluster is relatively compact and well-defined along the positive side of the first principal component (PC1), while the remaining clusters are more dispersed, with some overlap observed in the central region. This suggests that, although the clustering captures meaningful structure in the data, the boundaries between groups are not entirely distinct, reflecting a degree of continuous variation rather than fully discrete categories.

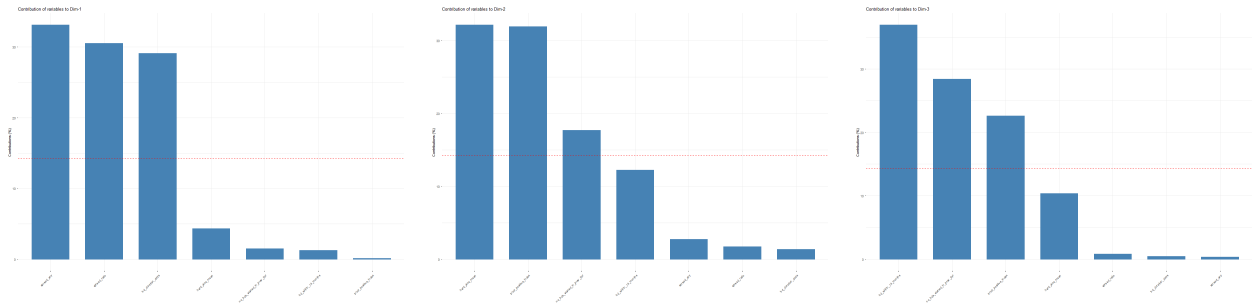


Figure 15: Contribution of variables to the first three principal components (PC1, PC2, and PC3).

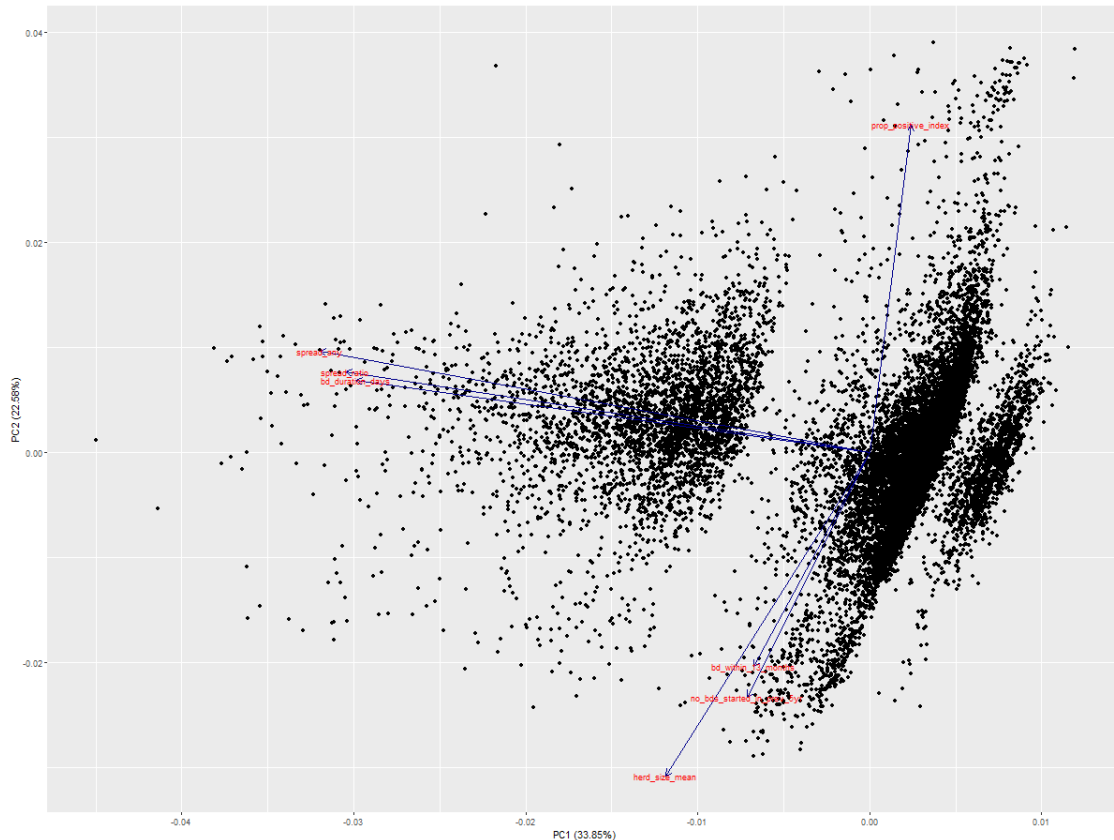


Figure 16: PCA biplot showing the relationship between observations and variable loadings on the first two principal components.

Figure 16 presents the PCA biplot, showing the relationships between variables and the principal components. Variables associated with transmission intensity, such as spread measures, are aligned along the negative direction of PC1, while variables related to herd size and recurrence contribute more strongly in the positive direction. The orientation of the vectors indicates that PC1 primarily captures a gradient from low to high transmission dynamics, whereas PC2 reflects secondary variation related to differences in recurrence and herd characteristics.

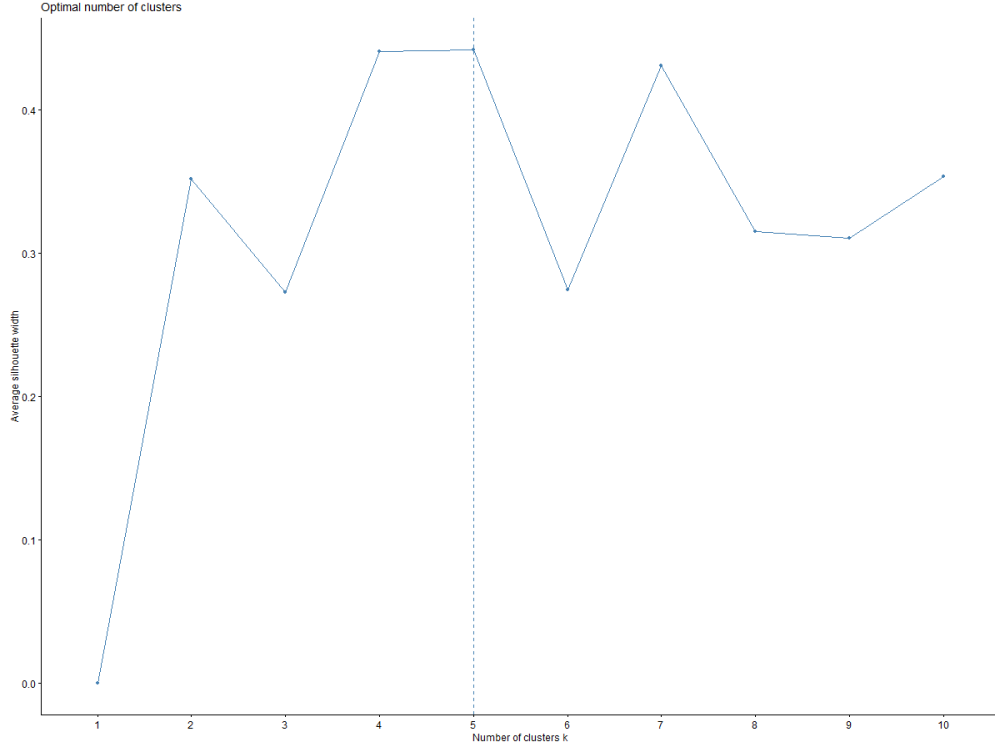


Figure 17: Average silhouette width for $K = 2$ to $K = 10$ based on k-means clustering.

Silhouette method for optimal K (Figure 17). The average silhouette width increases markedly from $K = 2$ and reaches a plateau around $K = 4-5$, indicating the presence of clustering structure. However, improvements beyond $K = 3$ are modest, suggesting limited gains in separation with additional clusters.

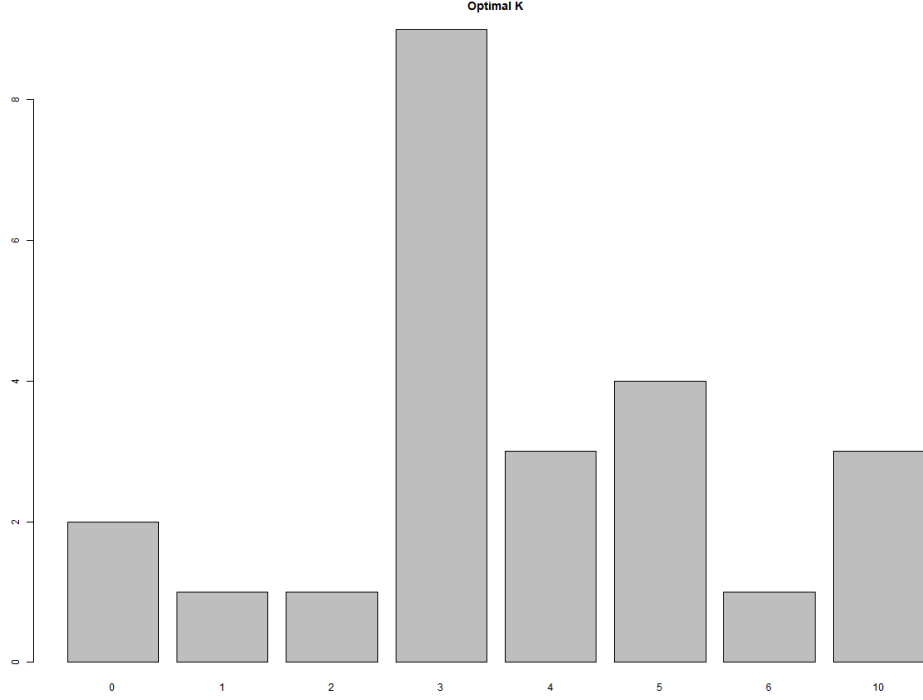


Figure 18: Number of clustering indices supporting each value of K from the NbClust analysis.

The results indicate that $K = 3$ is the most frequently recommended solution across the evaluated indices, receiving the highest level of support. While alternative values such as $K = 4$ and $K = 5$ are suggested by a smaller number of indices, their support is notably weaker.

This consensus across multiple independent criteria provides strong evidence that a three-cluster solution best captures the underlying structure of the data.

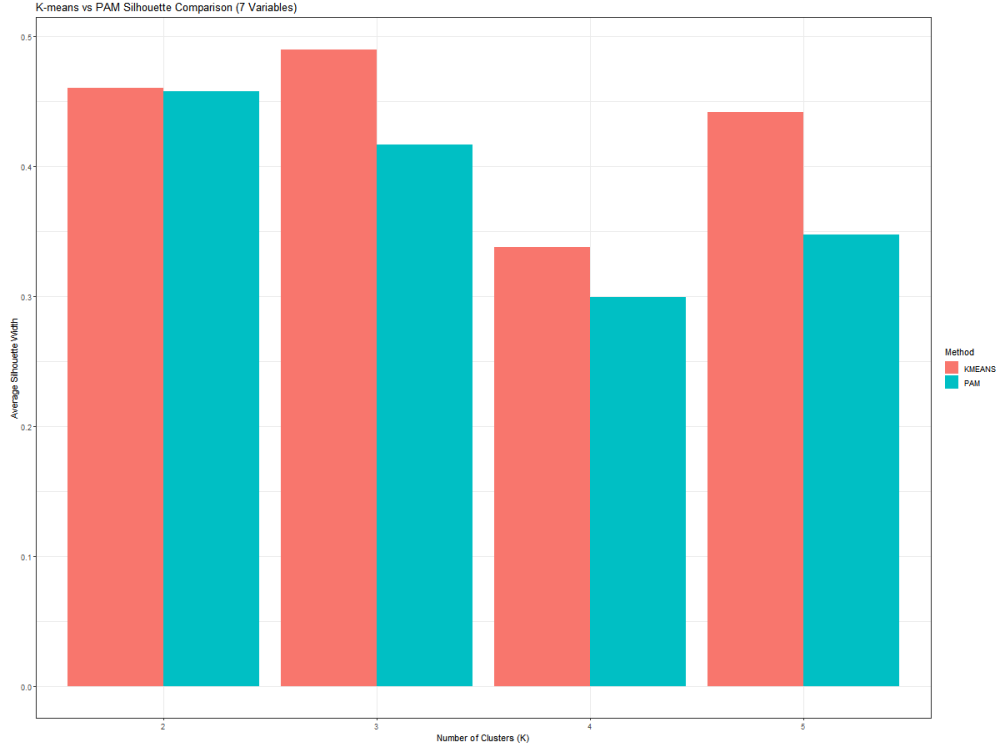


Figure 19: Comparison of average silhouette widths for K-means and PAM across different values of K .

K-means vs PAM comparison (Figure 19). K-means consistently achieves higher average silhouette widths than PAM across all values of K , indicating stronger cluster separation. The difference is most evident at $K = 3$, supporting its selection as a balanced and interpretable clustering solution.

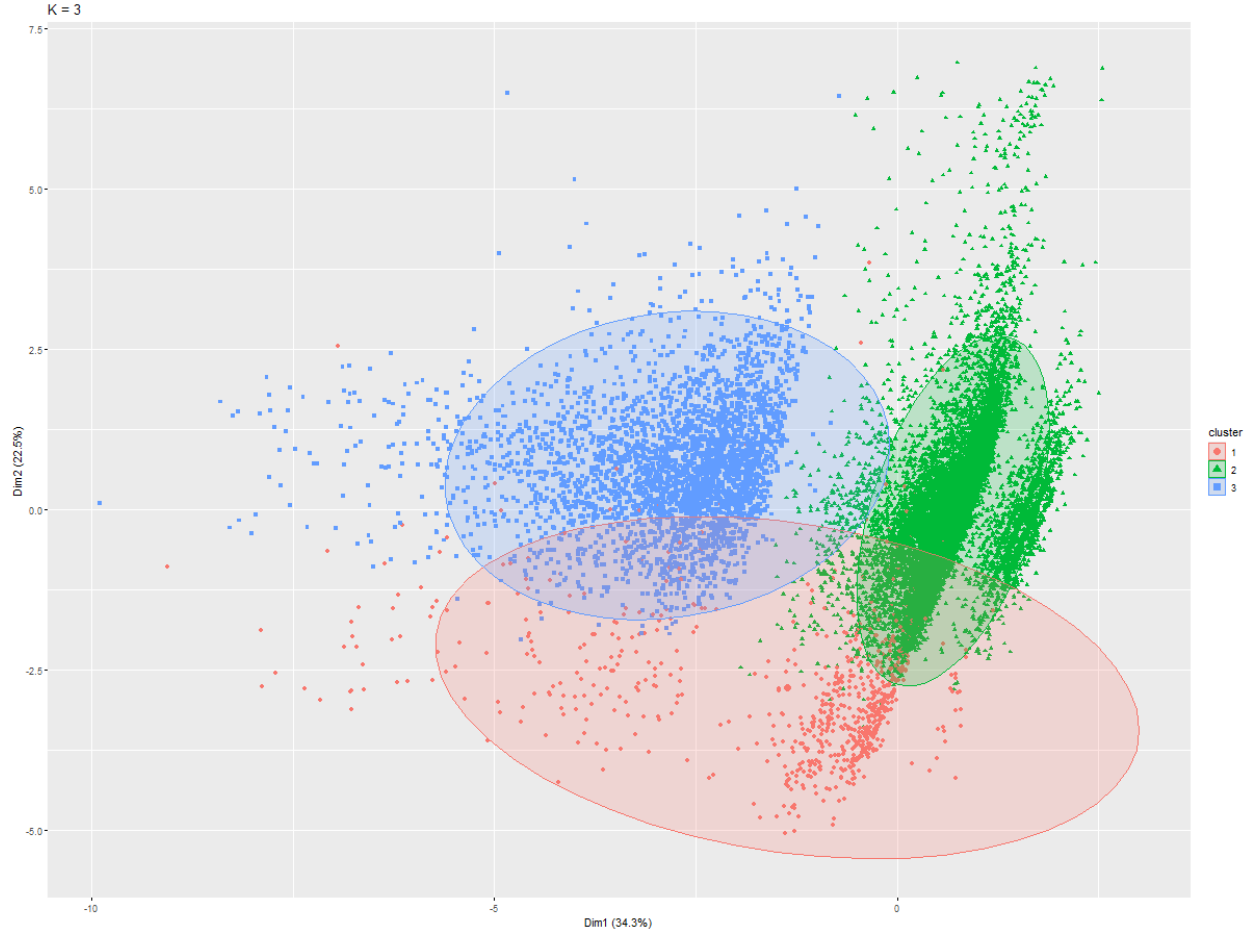


Figure 20: K-means clustering results for $K = 3$. Clusters are visualised using elliptical contours for clarity.

K-means clustering ($K = 3$) in PCA space (Figure 20). The three-cluster solution shows moderate separation in the PCA space. One cluster is compact and well-defined, while the others exhibit greater dispersion and partial overlap, suggesting meaningful but not perfectly distinct group structure.

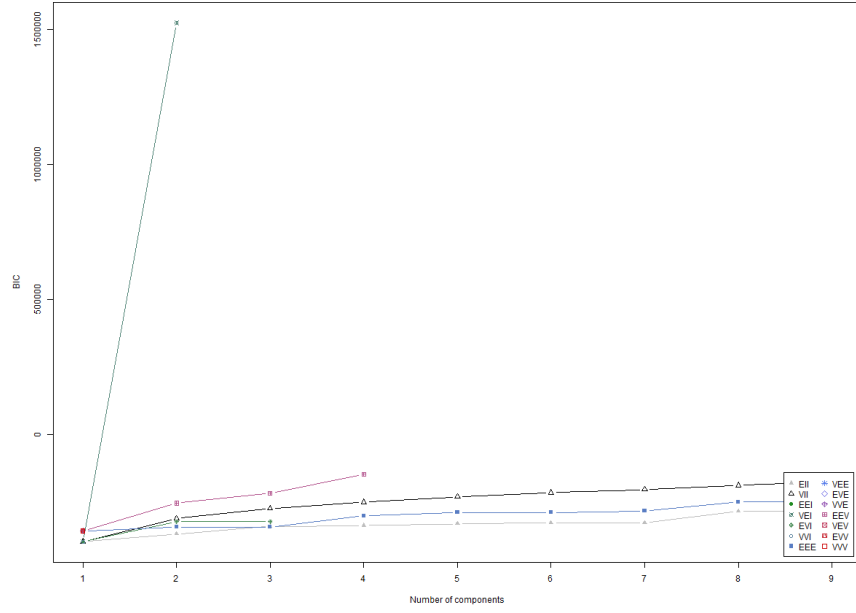


Figure 21: BIC values for GMM models with varying numbers of components. The optimal solution corresponds to $K = 2$.

Model-based clustering (BIC) (Figure 21). The BIC values indicate a clear preference for a low number of components, with the optimal model achieved at a small K . This suggests that the underlying structure of the data can be adequately captured with a limited number of clusters.

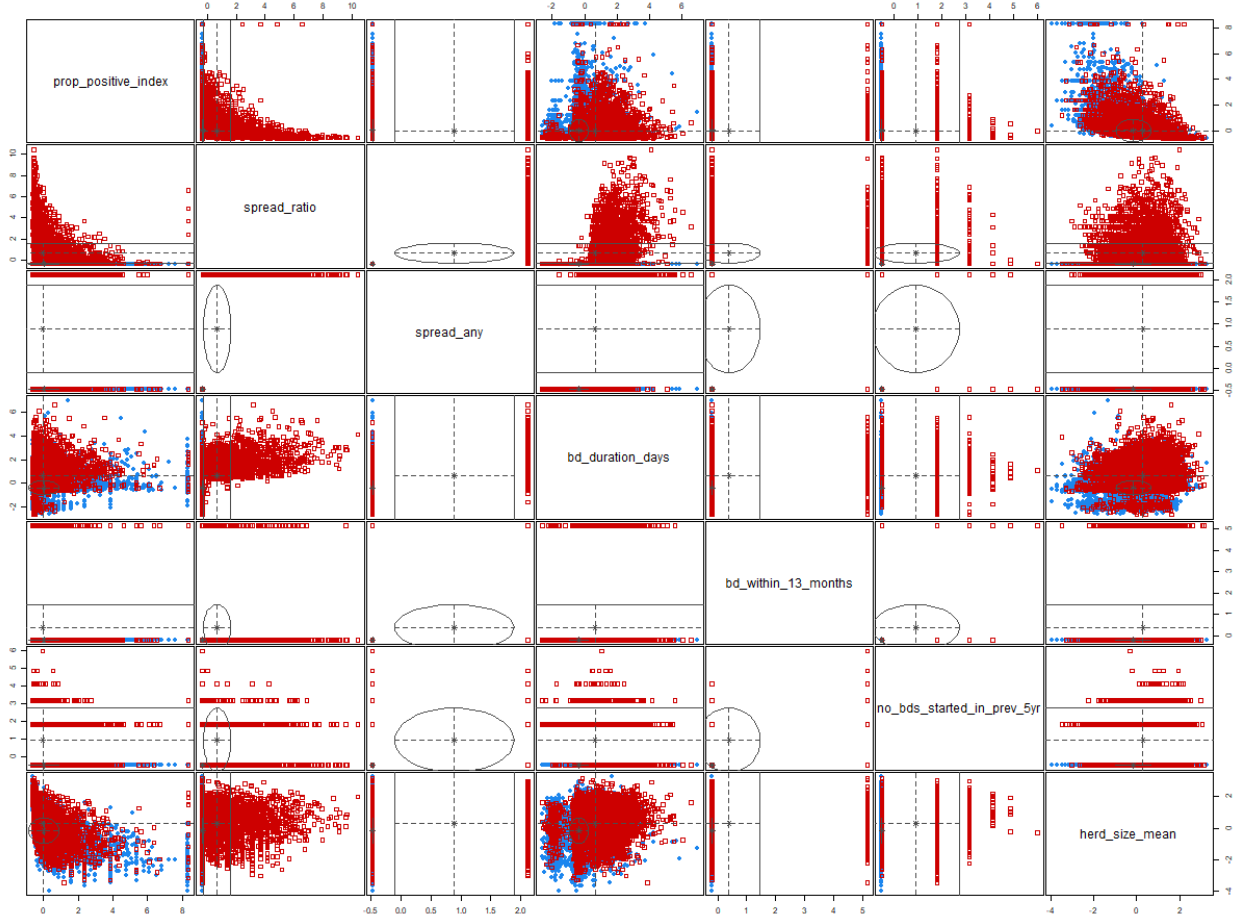


Figure 22: GMM clustering classification showing two-component structure and overlap between clusters.

GMM classification (Figure 22). The model-based clustering shows overlapping group assignments, indicating that cluster boundaries are not sharply defined. This reflects underlying heterogeneity and gradual transitions between groups rather than strictly separated clusters.

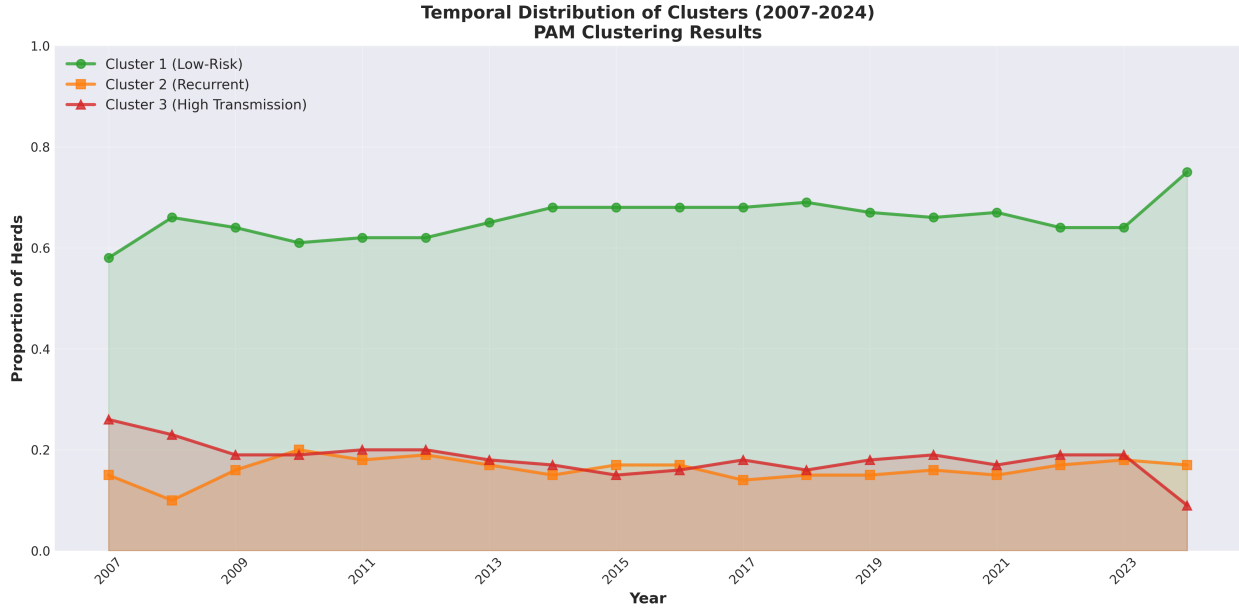


Figure 23: Temporal distribution of PAM clusters from 2007 to 2024, showing the proportion of herds assigned to each cluster over time.

Figure 23 illustrates the temporal distribution of clusters across the study period. Cluster 1 consistently represents the majority of herds, indicating that low-risk breakdown profiles dominate over time. Clusters 2 and 3 remain relatively stable but at lower proportions, with modest fluctuations across years. Notably, Cluster 3 shows a decline in the most recent year, while Cluster 1 increases, suggesting a potential shift toward less severe breakdown dynamics. Overall, the cluster structure remains broadly stable over time, supporting the robustness of the identified typologies.

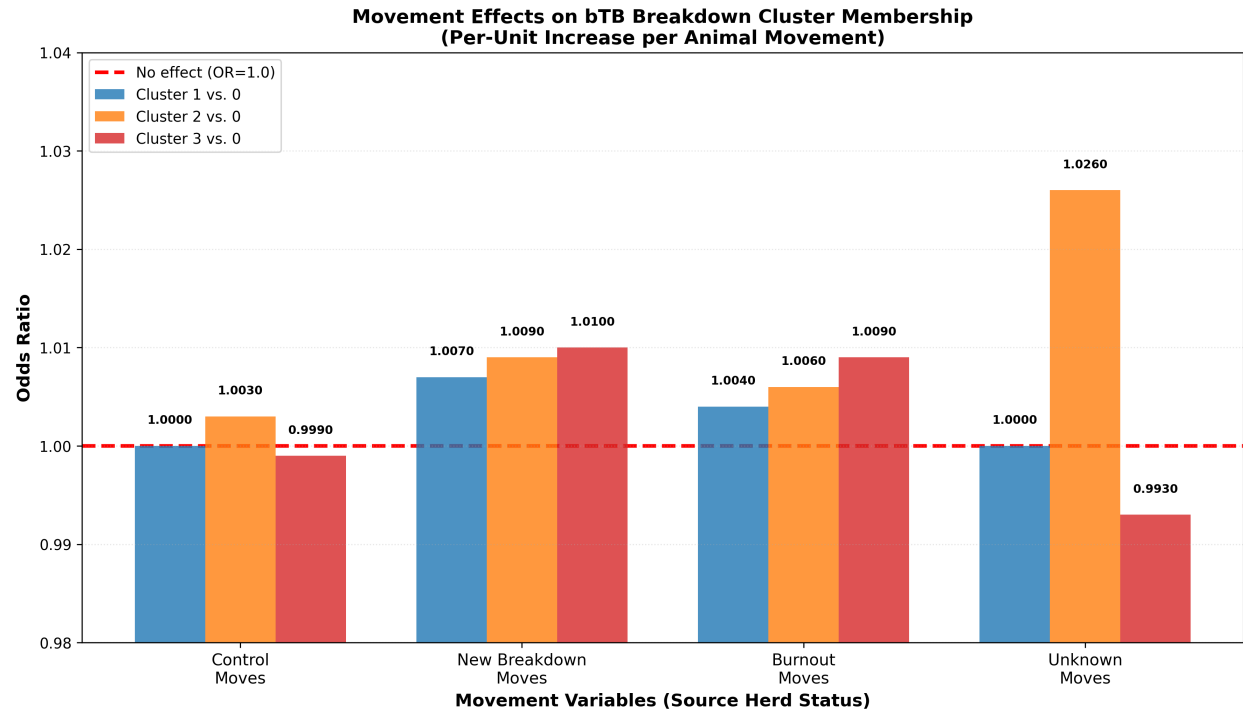


Figure 24: Movement effects on bTB breakdown cluster membership.

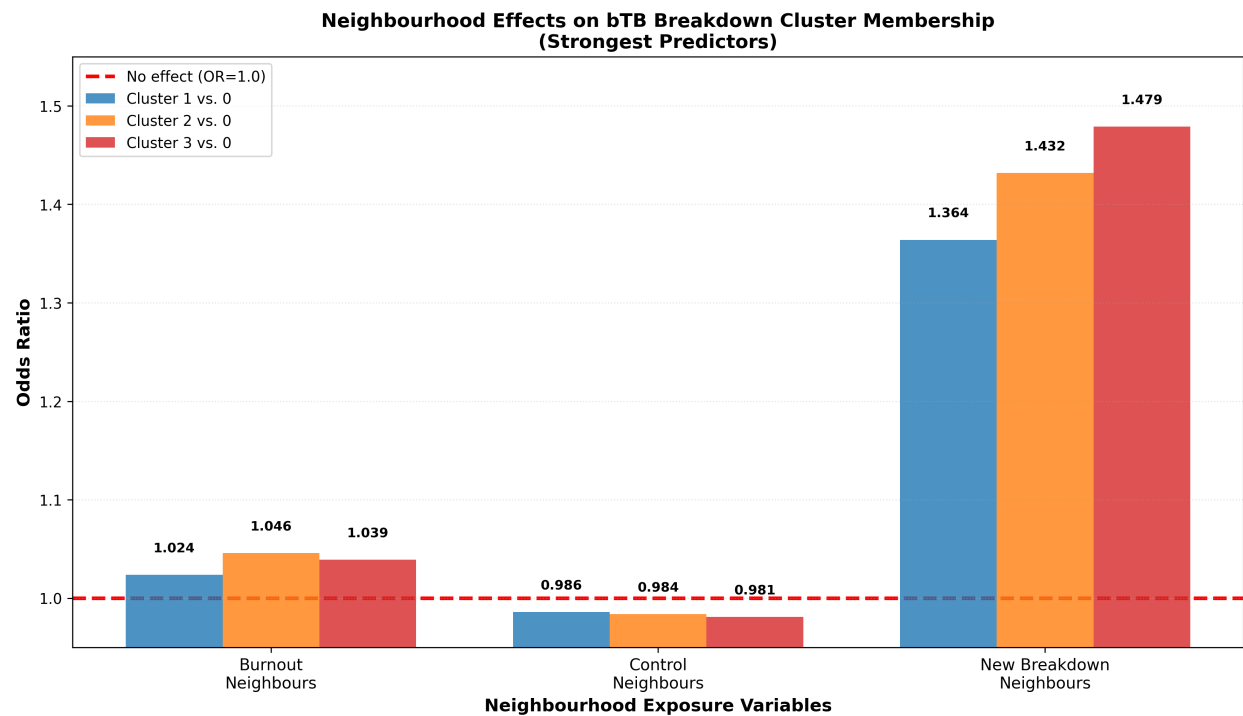


Figure 25: Neighbourhood effects on bTB breakdown cluster membership.

5.0.1. Kaplan–Meier survival curves

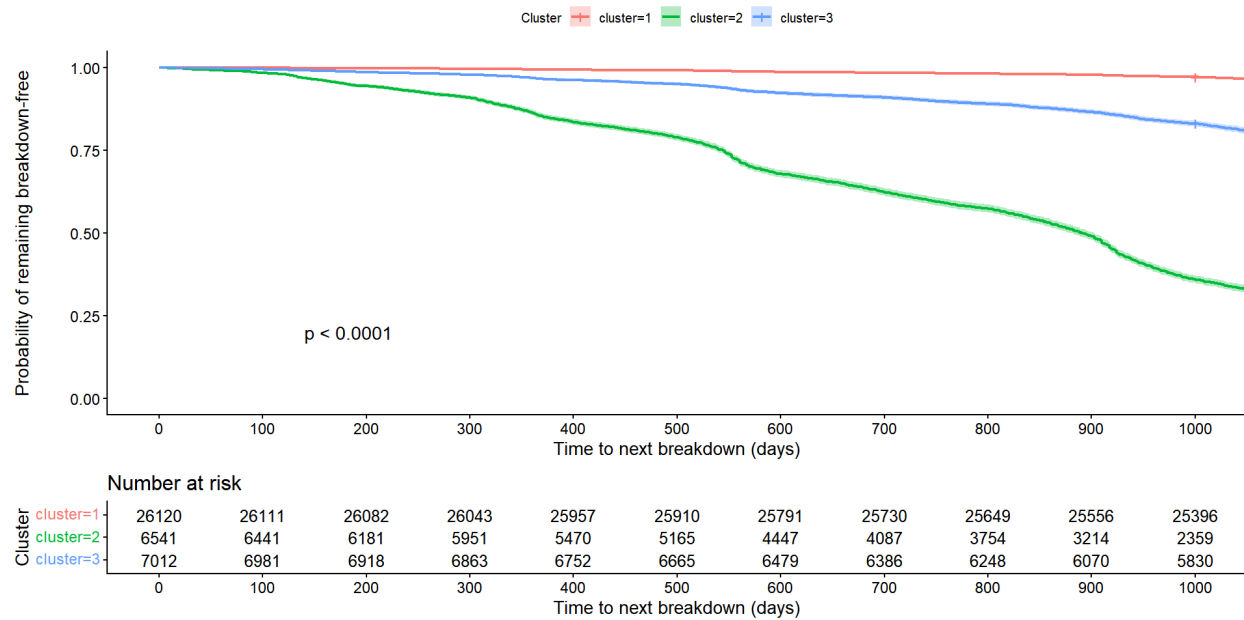


Figure 26: Kaplan–Meier curves illustrating time to subsequent breakdown by cluster. Shaded bands denote 95% confidence intervals. The risk table reports the number of herds remaining under observation at selected time points.